

Week 1 : Multivariable Calculus I

17. Find the first order partial derivatives with respect to x , y , z of the vector field $\mathbf{F}$:

(a) $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + 0\mathbf{k}$

(b) $\mathbf{F}(x, y, z) = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$

a) $\frac{\partial \mathbf{F}}{\partial x} = \begin{bmatrix} y \\ 0 \\ 0 \end{bmatrix}, \frac{\partial \mathbf{F}}{\partial y} = \begin{bmatrix} x \\ z \\ 0 \end{bmatrix}, \frac{\partial \mathbf{F}}{\partial z} = \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix}$

b) $\frac{\partial \mathbf{F}}{\partial x} = \begin{bmatrix} 0 \\ 0 \\ 2x \end{bmatrix}, \frac{\partial \mathbf{F}}{\partial y} = \begin{bmatrix} 2y \\ 0 \\ 0 \end{bmatrix}, \frac{\partial \mathbf{F}}{\partial z} = \begin{bmatrix} 0 \\ 2z \\ 0 \end{bmatrix}$

18. Consider the elliptic coordinate system (u, v) related to the Cartesian coordinates (x, y) by

$$x = \cosh(u) \cos(v), \quad y = \sinh(u) \sin(v) \quad \text{for } (u, v) \in [0, \infty) \times [0, 2\pi)$$

(a) Sketch in the xy -plane the curves given $u = 0, u = 1, u = 2$. Then sketch in the xy -plane the curves given $v = \frac{n\pi}{6}$ for $n = 0, 1, \dots, 11$.

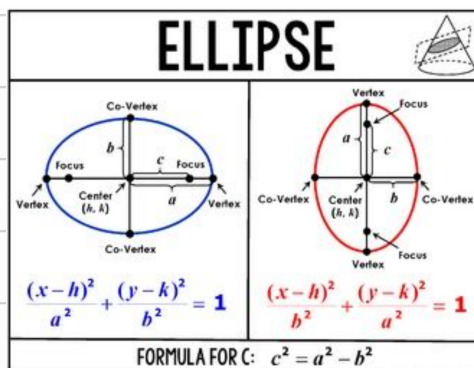
(b) Find the Jacobian for this change in coordinate system.

a) $\cos(v) = \frac{x}{\cosh(u)} \quad ; \quad \sin(v) = \frac{y}{\sinh(u)}$

If u constant, we eliminate variable v

Using $\cos^2(v) + \sin^2(v) = 1$

$$\frac{x^2}{\cosh^2(u)} + \frac{y^2}{\sinh^2(u)} = 1$$



Since $\cosh^2(u) > \sinh^2(u)$,

$\Rightarrow$ Horizontal Ellipse

- Semi-major axis, $a = \cosh(u)$
 - Semi-minor axis, $b = \sinh(u)$
 - $c^2 = a^2 - b^2$
- $$= \cosh^2(u) - \sinh^2(u)$$

$$c^2 = 1$$

$$c = \pm 1$$

$$\Rightarrow \text{foci} = (\pm 1, 0)$$

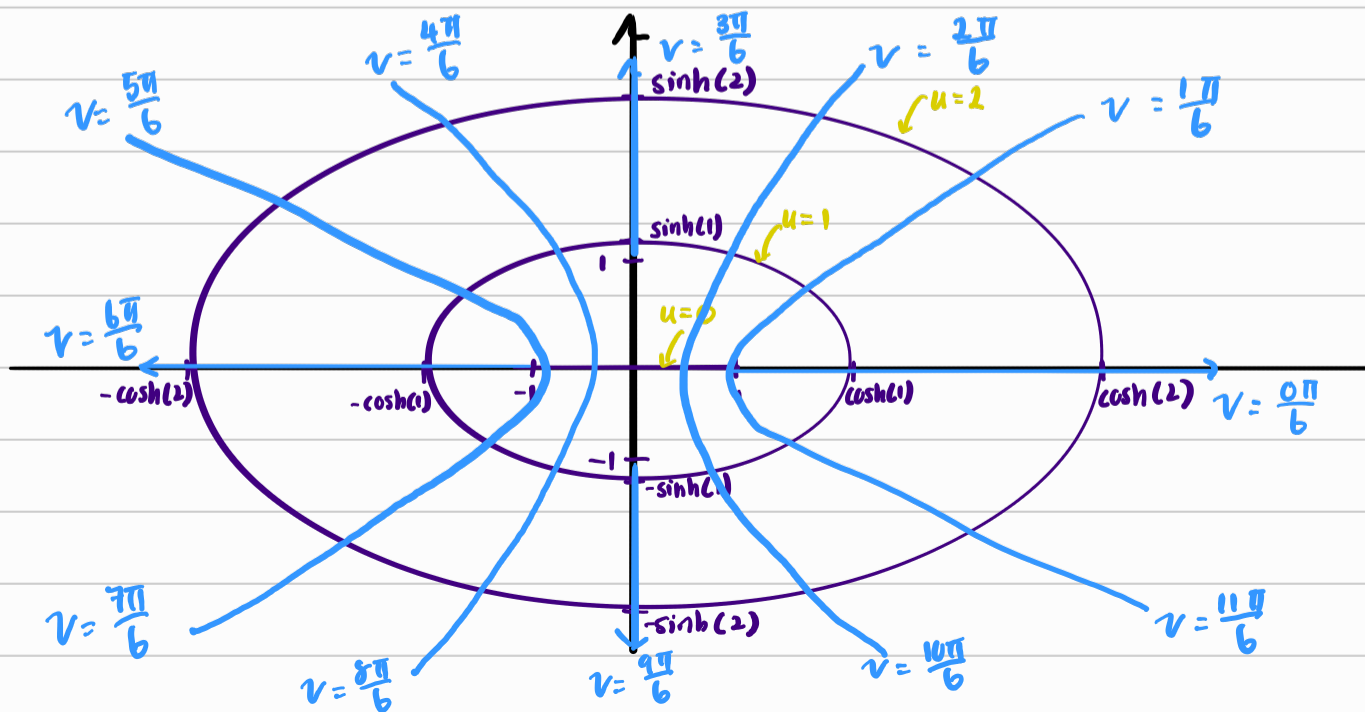
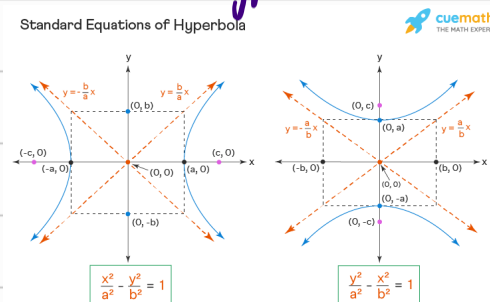
If v constant we eliminate variable u

Using $\cosh^2(u) - \sinh^2(u) = 1$

$$\frac{x^2}{\cosh^2(v)} - \frac{y^2}{\sinh^2(v)} = 1$$

Since x term > 0

$\Rightarrow$ Horizontal hyperbola



$$b) \quad J = \det \begin{bmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{bmatrix}$$

$$= \det \begin{bmatrix} \sinh(u) \cos(v) & -\cosh(u) \sin(v) \\ \cosh(u) \sin(v) & \sinh(u) \cos(v) \end{bmatrix}$$

$$= [\sinh(u) \cos(v)] [\sinh(u) \cos(v)] - [-\cosh(u) \sin(v)] [\cosh(u) \sin(v)]$$

$$= \sinh^2(u) \cos^2(v) + \cosh^2(u) \sin^2(v)$$

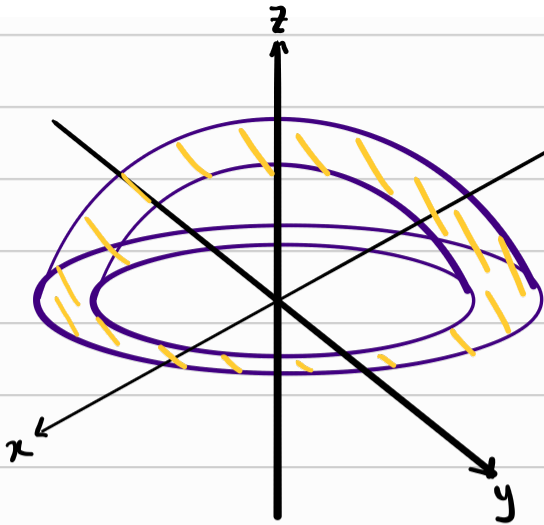
$$= \sinh^2(u) \cos^2(v) + [1 + \sinh^2(u)] \sin^2(v)$$

$$= \sinh^2(u) [\cos^2(v) + \sin^2(v)] + \sin^2(v)$$

$$= \sinh^2(u) + \sin^2(v) \quad \#$$

Week 2 : Multivariable Calculus II

14. Use triple integration to find the volume and mass of the three dimensional region above $z = 0$ and between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$, if the density is $\rho(x, y, z) = x^2 + y^2 + z^2$. $r=1$ $r=2$



By using spherical coordinates,
the boundaries are ...

$$1 \leq r \leq 2$$

$$0 \leq \phi \leq \pi/2$$

$$0 \leq \theta \leq 2\pi$$

$$\text{Volume} = \iiint 1 \, dV$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \int_1^2 1 [r^2 \sin(\phi) \, dr \, d\phi \, d\theta]$$

$$= \left[\int_0^{2\pi} d\theta \right] \left[\int_0^{\pi/2} \sin(\phi) \, d\phi \right] \left[\int_1^2 r^2 \, dr \right]$$

$$= [2\pi] [-\cos \phi]_0^{\pi/2} \left[\frac{r^3}{3} \right]_1^2$$

$$= (2\pi)(1)(7/3)$$

$$= 14\pi/3 \quad \#$$

$$\text{Mass} = \iiint \rho \, dV$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \int_1^2 (x^2 + y^2 + z^2) [r^2 \sin(\phi) \, dr \, d\phi \, d\theta]$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \int_1^2 (r^2) (r^2 \sin \phi) \, dr \, d\phi \, d\theta$$

$$= \left[\int_0^{2\pi} d\theta \right] \left[\int_0^{\pi/2} \sin \phi \, d\phi \right] \left[\int_1^2 r^4 \, dr \right]$$

$$= (2\pi)(1) \left[\frac{r^5}{5} \right]_1^2$$

$$= (2\pi)(1)(31/5)$$

$$= 62\pi/5 \quad \#$$

15. Consider the region $\mathcal{R}$ bounded by the lines $y = x + 2$ and $y = 5 - (x - 1)^2$

$$= 5 - [x^2 - 2x + 1]$$

$$= -x^2 + 2x + 4$$

- Sketch the region $\mathcal{R}$ in the xy -plane.
- Find the Cartesian coordinates of the two points where these curves intersect.
- Express the double integral (or integrals) that defines the area of the region $\mathcal{R}$ where the inner integration is over the x -variable. (do not solve)
- Express the double integral (or integrals) that defines the area of the region $\mathcal{R}$ where the inner integration is over the y -variable. (do not solve)
- Evaluate the integral from part (d) to find the area of the region $\mathcal{R}$.

a) Intersection : $x + 2 = -x^2 + 2x + 4$

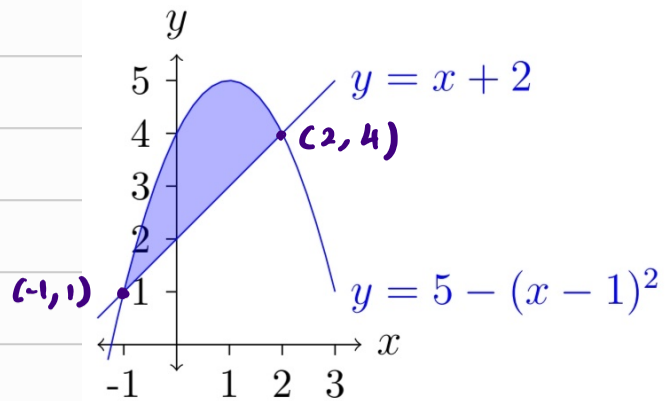
$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x - 2 \text{ or } x = -1$$

$$\Rightarrow y = 4 \quad \Rightarrow y = 1$$

$\therefore$ Intersection points : $(-1, 1), (2, 4)$



b) Answered in a)

c) $y = 5 - (x - 1)^2$

$$(x - 1)^2 = 5 - y$$

$$x = 1 \pm \sqrt{5 - y}$$

Lower Region : $1 - \sqrt{5 - y} \leq x \leq y - 1$

$$1 \leq y \leq 4$$

$$\therefore R_{\text{lower}} = \int_1^4 \int_{1 - \sqrt{5 - y}}^{y - 1} 1 \, dx \, dy$$

Upper Region : $1 - \sqrt{5 - y} \leq x \leq 1 + \sqrt{5 - y}$

$$4 \leq y \leq 5$$

$$\therefore R_{\text{upper}} = \int_4^5 \int_{1 - \sqrt{5 - y}}^{1 + \sqrt{5 - y}} 1 \, dx \, dy$$

$$\therefore R = \int_1^4 \int_{1 - \sqrt{5 - y}}^{y - 1} 1 \, dx \, dy + \int_4^5 \int_{1 - \sqrt{5 - y}}^{1 + \sqrt{5 - y}} 1 \, dx \, dy$$

d) $5 - (x - 1)^2 \leq y \leq x + 2$

$$-1 \leq x \leq 2$$

$$\therefore R = \int_{-1}^2 \int_{x + 2}^{5 - (x - 1)^2} 1 \, dy \, dx$$

$$e) R = \int_{-1}^2 \int_{5-(x-1)^2}^{x+2} 1 \, dy \, dx$$

$$= \int_{-1}^2 \left[y \right]_{5-(x-1)^2}^{x+2} dx$$

$$= \int_{-1}^2 \left[(-x^2 + 2x + 4) - (x + 2) \right] dx$$

$$= \int_{-1}^2 -x^2 + x + 2 \, dx$$

$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \frac{16}{3} - \left(-\frac{7}{6} \right)$$

$$= 4.5 \, *$$

16. Using spherical coordinates, evaluate $\iiint_V (x^2 + y^2 + z^2) \, dx \, dy \, dz$ where V is the part of the sphere having the centre at the origin, radius a and sits in the octant $x \leq 0, y \geq 0, z \geq 0$.

Boundaries

$$0 \leq r \leq a$$

$$0 \leq \phi \leq \pi/2$$

$$\pi/2 \leq \theta \leq \pi$$

$$\therefore \int_{\pi/2}^{\pi} \int_0^{\pi/2} \int_0^a (r^2) [r^2 \sin(\phi) \, dr \, d\phi \, d\theta]$$

$$= \left[\int_{\pi/2}^{\pi} d\theta \right] \left[\int_0^{\pi/2} \sin(\phi) \, d\phi \right] \left[\int_0^a r^4 \, d\theta \right]$$

$$= \left[\theta \right]_{\pi/2}^{\pi} \left[-\cos(\phi) \right]_0^{\pi/2} \left[\frac{r^5}{5} \right]_0^a$$

$$= (\pi/2) (1) (a^5/5)$$

$$= \frac{\pi a^5}{10} \quad \#$$

Week 3 : Vector Calculus I

13. Find the field lines of the three-dimensional vector functions

(a) $\mathbf{F}(x, y, z) = xy\mathbf{i} + (y^2 + 1)\mathbf{j} + z\mathbf{k}$.

(b) $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$.

(c) $\mathbf{F}(x, y, z) = y\mathbf{i} - x\mathbf{j} + \mathbf{k}$.

a) Let field line, $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

$$\vec{F}(\vec{r}(t)) = \vec{r}'(t)$$

$$\begin{bmatrix} xy \\ y^2 + 1 \\ z \end{bmatrix} = \begin{bmatrix} x'(t) \\ y'(t) \\ z'(t) \end{bmatrix}$$

$$\frac{dx}{dt} = xy \quad \text{--- ①}$$

$$\frac{dy}{dt} = y^2 + 1 \quad \text{--- ②}$$

$$\frac{dz}{dt} = z \quad \text{--- ③}$$

Solving for ②

$$\frac{dy}{dt} = y^2 + 1$$

$$\int \frac{1}{y^2 + 1} dy = \int 1 dt$$

$$\tan^{-1}(y) = t + B$$

$$y = \tan(t + B)$$

Solving for ③

$$\frac{dz}{dt} = z$$

$$\int \frac{1}{z} dz = \int 1 dt$$

$$\ln|z| = t + C_3$$

$$z(t) = e^{C_3} e^t$$

$$= C e^t$$

Solving for ①

$$\frac{dx}{dt} = xy$$

$$= x [\tan(t + C_2)]$$

$$\int \frac{1}{x} dx = \int \tan(t + C_2) dt$$

$$\ln|x| = \ln|\sec(t + C_2)| + C_1$$

$$x(t) = A \sec(t + C_2)$$

$$\therefore \mathbf{r}(t) = \begin{bmatrix} A \sec(t + C_2) \\ \tan(t + B) \\ C e^t \end{bmatrix} \#$$

b) Let field line $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

$$\vec{F}(\vec{r}(t)) = \vec{r}'(t)$$

$$\begin{bmatrix} yz \\ xz \\ xy \end{bmatrix} = \begin{bmatrix} x'(t) \\ y'(t) \\ z'(t) \end{bmatrix}$$

$$\frac{dx}{dt} = yz \Rightarrow x \frac{dx}{dt} = xyz \quad - (1)$$

$$\frac{dy}{dt} = xz \Rightarrow y \frac{dy}{dt} = xyz \quad - (2)$$

$$\frac{dz}{dt} = xy \Rightarrow z \frac{dz}{dt} = xyz \quad - (3)$$

$$(1) = (2)$$

$$x \frac{dx}{dt} = y \frac{dy}{dt}$$

$$\int x dx = \int y dy$$

$$\frac{x^2}{2} = \frac{y^2}{2} + C_1$$

$$x^2 - y^2 = C_1 \quad - (4)$$

$$y = \pm \sqrt{x^2 - C_1}$$

$$(1) = (3)$$

$$x \frac{dx}{dt} = z \frac{dz}{dt}$$

$$\int x dx = \int z dz$$

$$\frac{x^2}{2} = \frac{z^2}{2} + C_2$$

$$x^2 - z^2 = C_2 \quad - (5)$$

$$z = \pm \sqrt{x^2 - C_2}$$

$$\therefore \frac{dx}{dt} = (\sqrt{x^2 - C_1})(\sqrt{x^2 - C_2})$$

$$\int \frac{1}{\sqrt{x^2 - C_1} \sqrt{x^2 - C_2}} dx = \int 1 dt$$

[No way I'm doing this]

$\therefore$ The solution is actually the intersection of system of eqn (1), (2), (3)

We managed to simplify a system of 3 eqn to a system of 2 eqn [Eqn (4) & (5)].

So similarly, I can say the solution is the intersection of $x^2 - y^2 = C_1$ and $x^2 - z^2 = C_2$ #

c) Let field line $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

$$\vec{F}(\vec{r}(t)) = \vec{r}'(t)$$
$$\begin{bmatrix} y \\ -x \\ 1 \end{bmatrix} = \begin{bmatrix} x'(t) \\ y'(t) \\ z'(t) \end{bmatrix}$$

$$\frac{dx}{dt} = y \quad \text{--- ①}$$

$$\frac{dy}{dt} = -x \quad \text{--- ②}$$

$$\frac{dz}{dt} = 1 \Rightarrow z = t + C_3 \quad \text{--- ③}$$

Solving for ①

$$\frac{d^2x}{dt^2} = \frac{dy}{dx}$$

$$= -x$$

[using ②]

$$\frac{d^2x}{dt^2} + x = 0$$

Characteristic Egn : $\lambda^2 + 1 = 0$

$$\lambda = \pm i$$

$$\therefore \text{GS : } x(t) = A \cos(t) + B \sin(t)$$

$$\therefore y = \frac{dx}{dt}$$

$$= -A \sin(t) + B \cos(t)$$

$$\therefore \vec{r}(t) = \begin{bmatrix} A \cos(t) + B \sin(t) \\ -A \sin(t) + B \cos(t) \\ t + C_3 \end{bmatrix} \quad \#$$

14. Use the gradient vector to find a normal vector to the following surfaces at the given point P

(a) $x + y + z = 0$, $P: (x, y, z) = (1, 1, -2)$.

(b) $z = x^2 + y^2$, $P: (x, y, z) = (2, 2, 8)$.

(c) $x^2 + 3y^2 + 2z^2 = 6$, $P: (x, y, z) = (2, 0, 1)$.

If we force $f(x, y, z) = \text{constant}$,

$$\Rightarrow \text{Directional Derivative, } D_{\hat{u}}f = \nabla f \cdot \hat{u}$$

$$0 = \nabla f \cdot \hat{u}$$

$$\therefore \nabla f \perp \hat{u}$$

$\therefore$ Since $\hat{u}$ represents the directions I can step while staying on the surface

$\Rightarrow \nabla f$ is a normal vector to that surface.

a) Let $f(x, y, z) = x + y + z$,

When $f(x, y, z) = \text{constant}$,

$$\nabla f = \text{Normal vector}$$

$$\text{Normal vector, } \vec{N} = \langle 1, 1, 1 \rangle$$

At $P(1, 1, -2)$,

$$\vec{N}|_{(1, 1, -2)} = \langle 1, 1, 1 \rangle *$$

b) Let $f(x, y, z) = x^2 + y^2 - z$

When $f(x, y, z) = \text{constant}$,

$$\nabla f = \text{Normal vector}$$

$$\text{Normal vector, } \vec{N} = \langle 2x, 2y, -1 \rangle$$

At $P(2, 2, 8)$

$$\vec{N}|_{(2, 2, 8)} = \langle 4, 4, -1 \rangle *$$

c) Let $f(x, y, z) = x^2 + 3y^2 + 2z^2 - 6$

When $f(x, y, z) = \text{constant}$,

$$\nabla f = \text{Normal vector}$$

$$\text{Normal vector, } \vec{N} = \langle 2x, 6y, 4z \rangle$$

At $P(2, 0, 1)$

$$\vec{N}|_{(2, 0, 1)} = \langle 4, 0, 4 \rangle *$$

Week 4 : Vector Calculus II

13. Consider the curve $\mathbf{r}(t) = (t+1)\mathbf{i} + (\frac{1}{2} - \frac{1}{2}t)\mathbf{j} + 0\mathbf{k}$ with $-1 \leq t \leq 1$ from Week 3 question 1 and its new parametrisation

$$\mathbf{r}(s) = 2s\mathbf{i} + (1-s)\mathbf{j} + 0\mathbf{k} \text{ with } 0 \leq s \leq 1.$$

- (a) Using this new parametric representation, calculate the length of the line segment.
- (b) Derive a new parametric representation, $\mathbf{r}_*(\ell)$, of this line segment using the arc length as the parametric variable.

$$\begin{aligned} \text{a) Length} &= \int_0^1 \sqrt{\left(\frac{d\mathbf{r}}{ds}\right)^2} ds \\ &= \int_0^1 \sqrt{(2)^2 + (-1)^2 + (0)^2} ds \\ &= \int_0^1 \sqrt{5} ds \\ &= [\sqrt{5} s]_0^1 \\ &= \sqrt{5} \quad \# \end{aligned}$$

b) We have to replace "s" with "l" in $\mathbf{r}(s)$
 $\therefore$ We have to represent "s" in terms of "l"

$$\begin{aligned} l(s) &= \int_0^s \sqrt{\left(\frac{d\mathbf{r}}{ds}\right)^2} ds \\ &= \sqrt{5} s \quad \Rightarrow \quad 0 \leq s \leq 1 \\ &\quad \quad \quad 0 \leq l \leq \sqrt{5} \end{aligned}$$

$$\therefore s = l/\sqrt{5}$$

$$\begin{aligned} \therefore \mathbf{r}_*(l) &= \langle 2[l/\sqrt{5}], 1 - [l/\sqrt{5}], 0 \rangle \\ &= \langle 2l/\sqrt{5}, 1 - l/\sqrt{5}, 0 \rangle, \quad 0 \leq l \leq \sqrt{5} \quad \# \end{aligned}$$

14. Evaluate the line integral $\int_{AB} (x+y)dx + xdy$ for two paths of integration:

a) AB is the line segment A(0,0) to B(1,1).

b) AB is the parabola $y = x^2$ from A(0,0) to B(1,1).

a) Let the line segment be $y = x$, $0 \leq x \leq 1$
 $dy = dx$

$$\begin{aligned}\int_{AB} [(x+y) dx + x dy] &= \int_0^1 [(x+x) dx + x dx] \\ &= \int_0^1 3x dx \\ &= 3 \left[\frac{x^2}{2} \right]_0^1 \\ &= \frac{3}{2} \quad \# \end{aligned}$$

b) $y = x^2 \Rightarrow \frac{dy}{dx} = 2x$, $0 \leq x \leq 1$
 $dy = 2x dx$

$$\begin{aligned}\int_{AB} [(x+y) dx + x dy] &= \int_0^1 [(x+x^2) dx + x(2x dx)] \\ &= \int_0^1 (x + 3x^2) dx \\ &= \left[\frac{x^2}{2} + x^3 \right]_0^1 \\ &= \left(\frac{1}{2} + 1 \right) - 0 \\ &= \frac{3}{2} \quad \# \end{aligned}$$

15. Show that the line integral from A(1,2) to B(4,5), $\int_{AB} (3x^2y + y)dx + (x^3 + x)dy$ is path independent and that this integral is 336.

$$\vec{F} = \langle 3x^2y + y, x^3 + x \rangle$$

$$\nabla \times \vec{F} = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} 3x^2y + y \\ x^3 + x \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -0 \\ -(0 & -0) \\ (3x^2 + 1) - (3x^2 + 1) \end{pmatrix}$$

$$= \vec{0}$$

$\therefore$ Conservative

Since its a conservative vector field,

$$\vec{F} = \nabla f$$
$$\begin{pmatrix} 3x^2y + y \\ x^3 + x \end{pmatrix} = \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \end{pmatrix}$$

$$\frac{\partial f}{\partial x} = 3x^2y + y$$

$$\frac{\partial f}{\partial y} = x^3 + x$$

$$f = x^3y + xy + C_1(y)$$

$$\frac{\partial f}{\partial y} = x^3 + x + \frac{dC_1}{dy}$$

$$x^3 + x + \frac{dC_1}{dy} = x^3 + x$$

$$\frac{dC_1}{dy} = 0$$

$$C_1(y) = C$$

$$\therefore f = x^3y + xy + C, C \in \mathbb{R}$$

$\therefore$ Line integral AB

$$= f(B) - f(A)$$

$$= f(4,5) - f(1,2)$$

$$= [4^3(5) + 4(5) + C] - [1^3(2) + 1(2) + C]$$

$$= 340 - 4$$

$$= 336$$

Week 5 : Vector Calculus III

3. Consider the surface defined by $z = h(1 - x^2 - y^2)$, where h is a strictly positive constant, between $z = 0$ and $z = h$.

(a) Sketch the surface.

(b) Using the two-dimensional parametric representation

$$\mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + h(1 - u^2 - v^2)\mathbf{k}$$

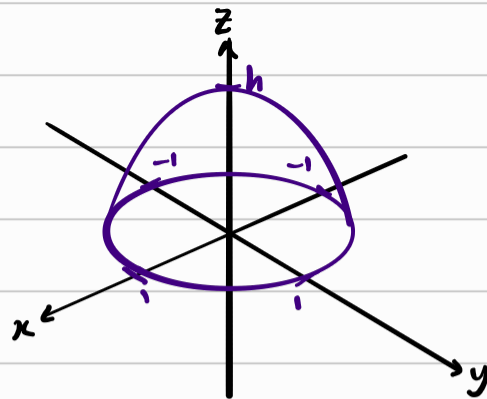
Define the bounds of integration for u and v , and set up the surface area integral but do not evaluate the integral.

(c) Using the parametric representation $\mathbf{r}(u, v) = u \cos(v)\mathbf{i} + u \sin(v)\mathbf{j} + h(1 - u^2)\mathbf{k}$:

i. Define the bounds of integration for u and v .

ii. Calculate the surface area.

a) Peak : when $x = 0, y = 0, z = h$
 Base : when $z = 0, 1 - x^2 - y^2 = 0$
 $x^2 + y^2 = 1$



b) Given $\mathbf{r}(u, v) = \langle u, v, 1 - u^2 - v^2 \rangle$

By comparison, $u = x, v = y$

Circular base : $u^2 + v^2 = 1$

$$v = \pm \sqrt{1 - u^2}$$

Boundaries : $-\sqrt{1 - u^2} \leq v \leq \sqrt{1 - u^2}$

$$-1 \leq u \leq 1$$

Surface Area, A

$$= \int_{-1}^1 \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} 1 \, dv \, du$$

Jacobian

$$= \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right|$$

$$= \left| \begin{bmatrix} 1 \\ 0 \\ -2hu \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ -2hv \end{bmatrix} \right|$$

$$= |\langle 2hu, 2hv, 1 \rangle|$$

$$= \sqrt{(2hu)^2 + (2hv)^2 + 1^2}$$

$$= \sqrt{4h^2(u^2 + v^2) + 1}$$

$$\therefore A = \int_{-1}^1 \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} \sqrt{4h^2(u^2 + v^2) + 1} \, dv \, du$$

c) i) Given $r(u, v) = \langle u \cos(v), u \sin(v), h(1 - u^2) \rangle$

By comparison, $x = u \cos(v)$, $y = u \sin(v)$

$\Rightarrow u$: radial distance, r

v : angle, θ

Boundaries

$$-1 \leq u \leq 1$$

$$0 \leq v \leq 2\pi$$

ii) Jacobian = $\left| \frac{\partial r}{\partial u} \times \frac{\partial r}{\partial v} \right|$

$$= \left| \begin{bmatrix} \cos(v) \\ \sin(v) \\ -2hu \end{bmatrix} \times \begin{bmatrix} -u \sin(v) \\ u \cos(v) \\ 0 \end{bmatrix} \right|$$

$$= \left| \langle 2hu^2 \cos(v), 2hu^2 \sin(v), u \rangle \right|$$

$$= \sqrt{(2hu^2 \cos(v))^2 + (2hu^2 \sin(v))^2 + u^2}$$

$$= \sqrt{4h^2 u^4 (\sin^2(v) + \cos^2(v)) + u^2}$$

$$= \sqrt{u^2 [4h^2 u^2 + 1]}$$

$$= u \sqrt{4h^2 u^2 + 1}$$

$$\therefore \text{Area} = \int_0^{2\pi} \int_0^1 (1) dS$$

$$= \int_0^{2\pi} \int_0^1 (1) [u \sqrt{4h^2 u^2 + 1}] du dv$$

$$= \left[\int_0^{2\pi} 1 dv \right] \left[\int_0^1 u \sqrt{4h^2 u^2 + 1} du \right]$$

$$= 2\pi \int_0^1 \sqrt{4h^2 u^2 + 1} \cdot u du$$

$$= 2\pi \int_1^{4h^2+1} \sqrt{w} \cdot \left[\frac{1}{8h^2} dw \right]$$

$$= (2\pi) \cdot \frac{1}{8h^2} \left[\frac{2}{3} w^{3/2} \right]_1^{4h^2+1}$$

$$= \frac{\pi}{6h^2} [w^{3/2}]_1^{4h^2+1}$$

$$= \frac{\pi}{6h^2} [(4h^2+1)^{3/2} - 1]$$

$$\text{let } w = 4h^2 u^2 + 1$$

$$\frac{dw}{du} = 8h^2 u$$

$$u du = \frac{1}{8h^2} dw$$

$$\text{Limits : } u = 0, w = 1$$

$$u = 1, w = 4h^2 + 1$$

Week 6 : Vector Calculus IV

7. Consider the vector field $\mathbf{F} = (x^2 + y^2 + z^2)\mathbf{i}$ and a curve $\mathcal{C}$ defined by $\mathbf{r}(\theta) = a \cos \theta \mathbf{i} + a \sin \theta \mathbf{j} + a\theta \mathbf{k}$ corresponding to $0 \leq \theta \leq \pi/2$.

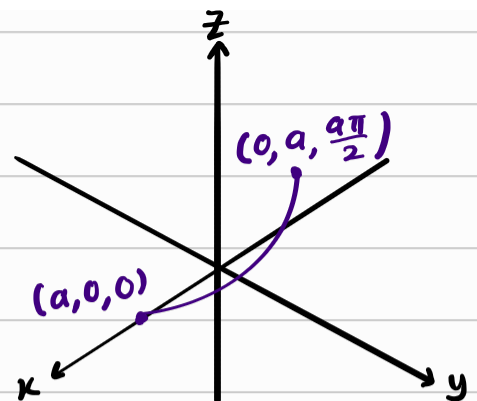
(a) Sketch the curve $\mathcal{C}$ and find its endpoints.

(b) Evaluate the vector field $\mathbf{F}$ on the curve $\mathcal{C}$.

(c) Calculate $\int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r}$. (Hint: You may need to do some integration by parts)

a) $\mathcal{C} : \mathbf{r}(\theta) = \langle a \cos \theta, a \sin \theta, a\theta \rangle$

Since $\begin{cases} x = a \cos \theta \\ y = a \sin \theta \end{cases} \left. \begin{array}{l} \text{circle} \\ \text{Helix} \end{array} \right\}$
 $z = a\theta$



b) $\vec{F}(\mathbf{r}(\theta)) = \vec{F}(a \cos \theta, a \sin \theta, a\theta)$
 $= \langle a^2 \cos^2 \theta + a^2 \sin^2 \theta + a^2 \theta^2, 0, 0 \rangle$
 $= \langle a^2 + a^2 \theta^2, 0, 0 \rangle$
 $= \langle a^2(1 + \theta^2), 0, 0 \rangle \neq$

c) Work Done $= \int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$
 $= \int_0^{\pi/2} \vec{F}(\mathbf{r}(\theta)) \cdot \frac{d\vec{r}}{d\theta} d\theta$
 $= \int_0^{\pi/2} \begin{bmatrix} a^2(1 + \theta^2) \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -a \sin \theta \\ a \cos \theta \\ a \end{bmatrix} d\theta$
 $= \int_0^{\pi/2} -a^3(1 + \theta^2) \sin \theta d\theta$
 $= -a^3 \int_0^{\pi/2} (1 + \theta^2) \sin \theta d\theta$
 $= -a^3 \left[-(1 + \theta^2) \cos \theta + 2\theta \sin \theta + 2 \cos \theta \right]_0^{\pi/2}$
 $= -a^3 \left[-\cos \theta - \theta^2 \cos \theta + 2 \cos \theta + 2\theta \sin \theta \right]_0^{\pi/2}$
 $= -a^3 \left[(1 - \theta^2) \cos \theta + 2\theta \sin \theta \right]_0^{\pi/2}$
 $= -a^3 \left[(\pi) - (1) \right]$
 $= a^3(1 - \pi) \neq$

D	I
$+ 1 + \theta^2$	$\sin \theta$
$- 2\theta$	$-\cos \theta$
$+ 2$	$-\sin \theta$
$- 0$	$\cos \theta$

8. Use the Divergence Theorem to compute the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$ of the vector field $\mathbf{F} = \cos(y)\mathbf{i} + \sin(x)\mathbf{j} + \cos(z)\mathbf{k}$, where S is the total surface area of the cylinder $x^2 + y^2 = 4$, $-2 \leq z \leq 3$ (including the ends) and $\hat{\mathbf{n}}$ is the outward normal.

Gauss Divergence Theorem : $\iint_{\text{Surface}} \vec{F} \cdot \hat{n} dS = \iiint_{\text{Volume}} \nabla \cdot \mathbf{F} dV$

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \cdot \begin{pmatrix} \cos(y) \\ \sin(x) \\ \cos(z) \end{pmatrix} \\ &= 0 + 0 - \sin(z) \\ &= -\sin(z) \end{aligned}$$

Boundaries using cylindrical coordinates

$$0 \leq r \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$-2 \leq z \leq 3$$

$$\begin{aligned} \therefore \text{Flux} &= \int_0^{2\pi} \int_0^2 \int_{-2}^3 -\sin(z) dV \\ &= \int_0^{2\pi} \int_0^2 \int_{-2}^3 -\sin(z) [r dz dr d\theta] \\ &= \left[\int_0^{2\pi} d\theta \right] \left[\int_0^2 r dr \right] \left[\int_{-2}^3 -\sin(z) dz \right] \\ &= [\theta]_0^{2\pi} \left[\frac{r^2}{2} \right]_0^2 [\cos(z)]_{-2}^3 \\ &= (2\pi) (2) (\cos(3) - \cos(-2)) \\ &= 4\pi [\cos(3) - \cos(2)] \quad \text{, since } \cos(x) = \cos(-x) \end{aligned}$$

Week 7 : Laplace Transform I

7. Find $\mathcal{L}^{-1}\{F(s)\}$ when $F(s)$ is given by $\frac{1}{(s+1)(s+2)(s^2+2s+10)}$.

$$\frac{1}{(s+1)(s+2)(s^2+2s+10)} = \frac{A}{(s+1)} + \frac{B}{(s+2)} + \frac{Cs+D}{s^2+2s+10}$$

$$1 = A(s+2)(s^2+2s+10) + B(s+1)(s^2+2s+10) + (Cs+D)(s+1)(s+2)$$

$$\text{When } s = -1 \Rightarrow A = \frac{1}{9}$$

$$s = -2 \Rightarrow B = -\frac{1}{10}$$

By matching s^3 terms of LHS & RHS

$$0 = A + B + C$$

$$= \frac{1}{9} - \frac{1}{10} + C$$

$$C = -\frac{1}{90}$$

$$\text{When } s = 0$$

$$1 = A(2)(10) + B(1)(10) + D(1)(2)$$

$$1 = 20A + 10B + 2D$$

$$1 = 20\left(\frac{1}{9}\right) + 10\left(-\frac{1}{10}\right) + 2D$$

$$D = -\frac{1}{9}$$

$$\therefore F(s) = \frac{1}{9} \left[\frac{1}{s+1} \right] - \frac{1}{10} \left[\frac{1}{s+2} \right] - \frac{1}{90} \left[\frac{s+10}{s^2+2s+10} \right]$$

$$f(t) = \frac{1}{9} e^{-t} - \frac{1}{10} e^{-2t} - \frac{1}{90} \mathcal{L}^{-1} \left\{ \frac{s+10}{s^2+2s+10} \right\} \quad \text{--- (1)}$$

By completing the square

$$s^2 + 2s + 10 = (s^2 + 2s + 1) + 9$$

$$= (s+1)^2 + 3^2$$

$$\therefore \frac{s+10}{s^2+2s+10} = \frac{(s+1)+9}{(s+1)^2+3^2} = \frac{s+1}{(s+1)^2+3^2} + 3 \left[\frac{3}{(s+1)^2+3^2} \right]$$

$$\begin{aligned} \therefore \mathcal{L}^{-1} \left\{ \frac{s+10}{s^2+2s+10} \right\} &= \mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2+3^2} + 3 \left[\frac{3}{(s+1)^2+3^2} \right] \right\} \\ &= e^{-t} \cos(3t) + 3e^{-t} \sin(3t) \quad \text{--- (2)} \end{aligned}$$

② $\rightarrow$ ①

$$\begin{aligned} f(t) &= \frac{1}{9} e^{-t} - \frac{1}{10} e^{-2t} - \frac{1}{90} \left[e^{-t} \cos(3t) + 3e^{-t} \sin(3t) \right] \\ &= \frac{1}{9} e^{-t} - \frac{1}{10} e^{-2t} - \frac{1}{90} e^{-t} \cos(3t) - \frac{1}{30} e^{-t} \sin(3t) \quad \# \end{aligned}$$

8. Using Laplace transform methods, solve for $t \geq 0$ the following differential equations, subject to the specified initial conditions:

(a) $\frac{d^3x}{dt^3} - 2\frac{d^2x}{dt^2} - \frac{dx}{dt} + 2x = 2 + t$, subject to $x = 0$ and $dx/dt = 1$, $\frac{d^2x}{dt^2} = 0$ at $t = 0$.

(b) $\frac{d^3x}{dt^3} + \frac{d^2x}{dt^2} + \frac{dx}{dt} + x = \cos(3t)$, subject to $x = 0$ and $dx/dt = 1$, $\frac{d^2x}{dt^2} = 1$ at $t = 0$.

Interpret the behaviour of your solution, for t very large. Which term in the solution will dominate? What is the homogeneous part of the solution?

a) $\mathcal{L}\{x\} = X(s)$

$\mathcal{L}\{x'\} = sX(s) - x(0) = sX(s)$

$\mathcal{L}\{x''\} = s^2X(s) - sx(0) - x'(0)$
 $= s^2X(s) - 1$

$\mathcal{L}\{x'''\} = s^3X(s) - s^2x(0) - sx'(0) - x''(0)$
 $= s^3X(s) - s$

$\frac{d^3x}{dt^3} - 2\frac{d^2x}{dt^2} - \frac{dx}{dt} + 2x = 2 + t$

Applying Laplace Transform to both sides

$\Rightarrow (s^3X(s) - s) - 2(s^2X(s) - 1) - (sX(s)) + 2X(s) = \mathcal{L}\{2 + t\}$

$X(s) [s^3 - 2s^2 - s + 2] - s + 2 = \frac{2}{s} + \frac{1}{s^2}$

$X(s) [s^3 - 2s^2 - s + 2] = \frac{2s + 1}{s^2} + s - 2$

$X(s) [(s^2 - 1)(s - 2)] = \frac{s^3 - 2s^2 + 2s + 1}{s^2}$

$X(s) [(s-1)(s+1)(s-2)] = \frac{s^3 - 2s^2 + 2s + 1}{s^2}$

These terms exist because $X(s) = \frac{s^3 - 2s^2 + 2s + 1}{s^2(s-1)(s+1)(s-2)}$ — (1)
 of $\frac{1}{s^2}$ that comes from $2 + t$

$\frac{s^3 - 2s^2 + 2s + 1}{s^2(s-1)(s+1)(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1} + \frac{D}{s+1} + \frac{E}{s-2}$ — (2)

$s^3 - 2s^2 + 2s + 1 = A(s)(s-1)(s+1)(s-2)$

$+ B(s-1)(s+1)(s-2)$

$+ C(s^2)(s+1)(s-2)$

$+ D(s^2)(s-1)(s-2)$

$+ E(s^2)(s-1)(s+1)$

$$\text{When } s = 0 \Rightarrow B = \frac{1}{(-1)(1)(-2)} = \frac{1}{2}$$

$$s = 1 \Rightarrow C = \frac{1-2+2+1}{(1^2)(1+1)(1-2)} = -1$$

$$s = -1 \Rightarrow D = \frac{-4}{(1)(-2)(-3)} = -\frac{2}{3}$$

$$s = 2 \Rightarrow E = \frac{5}{(4)(1)(3)} = \frac{5}{12}$$

$$\underline{\text{Let } s = 3}, \quad 16 = 24A + 8B + 36C + 18D + 72E$$

$$16 = 24A + 8(\frac{1}{2}) + 36(-1) + 18(-\frac{2}{3}) + 72(\frac{5}{12})$$

$$A = \frac{5}{4}$$

$$\therefore X(s) = \frac{5}{4} \left(\frac{1}{s} \right) + \frac{1}{2} \left(\frac{1}{s^2} \right) - \frac{1}{s-1} - \frac{2}{3} \left(\frac{1}{s+1} \right) + \frac{5}{12} \left(\frac{1}{s-2} \right)$$

$$x(t) = \frac{5}{4} + \frac{1}{2}t - e^{-t} - \frac{2}{3}e^{-t} + \frac{5}{12}e^{2t} \quad \#$$

$$\therefore \text{Dominating term} : \frac{5}{12}e^{2t} \quad \#$$

$\therefore$ Since driving func., $(2+t)$ created s^2 in the denominator of eqn ①, that generated these two terms in eqn ②, $\frac{A}{s}$ and $\frac{B}{s^2}$
 $\Rightarrow \frac{A}{s}$ and $\frac{B}{s^2}$ is due to the driving func., the rest is the

homogeneous part, $x_h(t)$

$$\Rightarrow x_h(t) = -e^{-t} - \frac{2}{3}e^{-t} + \frac{5}{12}e^{2t} \quad \#$$

$$b) \mathcal{L}\{x\} = X(s)$$

$$\mathcal{L}\{x'\} = sX(s) - \cancel{x(0)} = sX(s)$$

$$\mathcal{L}\{x''\} = s^2X(s) - s\cancel{x(0)} - \cancel{x'(0)}$$

$$= s^2X(s) - 1$$

$$\mathcal{L}\{x'''\} = s^3X(s) - s^2\cancel{x(0)} - s\cancel{x'(0)} - \cancel{x''(0)}$$

$$= s^3X(s) - s - 1$$

$$x''' + x'' + x' + x = \cos(3t)$$

Applying Laplace Transform on both sides,

$$(s^3X(s) - s - 1) + (s^2X(s) - 1) + (sX(s)) + X(s) = \mathcal{L}\{\cos(3t)\}$$

$$X(s) [s^3 + s^2 + s + 1] - s - 2 = \frac{s}{s^2 + 9}$$

$$X(s) [(s^2 + 1)(s + 1)] = \frac{s}{s^2 + 9} + s + 2$$

$$X(s) = \frac{s + (s+2)(s^2+9)}{(s^2+9)(s^2+1)(s+1)}$$

$$= \frac{s^3 + 2s^2 + 10s + 18}{(s+1)(s^2+1)(s^2+9)}$$

$$\frac{s^3 + 2s^2 + 10s + 18}{(s+1)(s^2+1)(s^2+9)} = \frac{A}{s+1} + \frac{Bs + C}{s^2+1} + \frac{Ds + E}{s^2+9} \quad \text{--- (1)}$$

$$s^3 + 2s^2 + 10s + 18 = A(s^2+1)(s^2+9) + (Bs+C)(s+1)(s^2+9) + (Ds+E)(s+1)(s^2+1)$$

$$+ (Bs + C)(s+1)(s^2+9)$$

$$+ (Ds + E)(s+1)(s^2+1)$$

$$\text{When } s = -1 \Rightarrow A = \frac{-1 + 2 - 10 + 18}{2(10)} = 9/20$$

$$\text{Let } s^2 = -1, s = i \quad [\text{To eliminate } (A) \& (Ds + E)]$$

$$\Rightarrow -i - 2 + 10i + 18 = A(0) + (Bi + C)(i+1)(-1+9) + (Ds + E)(0)$$

$$16 + 9i = 8(Bi + C)(i+1)$$

$$16 + 9i = 8[-B + Bi + Ci + C]$$

$$= 8[-B + C] + 8[B + C]i$$

By comparing real & imaginary parts

$$8(-B + C) = 16$$

$$8(B + C) = 9$$

$$\Rightarrow C - B = 2$$

$$\Rightarrow C + B = 9/8$$

$$(2 + B) + B = 9/8$$

$$B = -7/16$$

$$\therefore C = 25/16$$

Let $s^2 = -9$, $s = 3i$ [To eliminate (A) & (Bs + C)]

$$\Rightarrow -27i - 18 + 30i + 18 = (3Di + E)(3i + 1)(-9 + 1)$$

$$3i = -8(3Di + E)(3i + 1)$$

$$= -8[-9D + 3Di + 3Ei + E]$$

$$= -8[-9D + E] - 8[3D + 3E]i$$

By comparing real & imaginary part

$$-8[-9D + E] = 0$$

$$-8[3D + 3E] = 3$$

$$\Rightarrow E = 9D$$

$$\Rightarrow -8[3D + 3(9D)] = 3$$

$$30D = -3/8$$

$$D = -1/80$$

$$\therefore E = -9/80$$

A

Bs + C

Ds + E

$$\therefore X(s) = \frac{9}{20} \left(\frac{1}{s+1} \right) + \left[-\frac{7}{16} \left(\frac{s}{s^2+1} \right) + \frac{25}{16} \left(\frac{1}{s^2+1} \right) \right] + \left[-\frac{1}{80} \left(\frac{s}{s^2+9} \right) - \frac{9}{80} \left(\frac{1}{s^2+9} \right) \right]$$

$$= \frac{9}{20} \left(\frac{1}{s+1} \right) - \frac{7}{16} \left(\frac{s}{s^2+1} \right) + \frac{25}{16} \left(\frac{1}{s^2+1} \right) - \frac{1}{80} \left(\frac{s}{s^2+3^2} \right) - \frac{3}{80} \left(\frac{1}{s^2+3^2} \right)$$

$$x(t) = \frac{9}{20} e^{-t} - \frac{7}{16} \cos(3t) + \frac{25}{16} \sin(3t) - \frac{1}{80} \cos(3t) - \frac{3}{80} \sin(3t) \quad \#$$

$\therefore$ Dominating term : Since $e^{-t} \rightarrow 0$ as $t \rightarrow \infty$, that leaves behind sine & cosine.

Sine & cosine just oscillates btwn 1 and -1

$\Rightarrow$ No single term dominates. $\#$

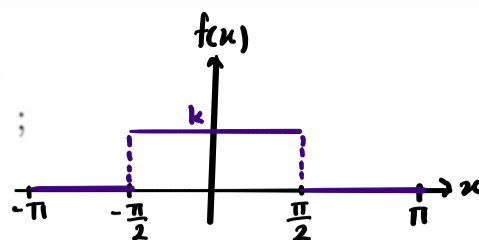
$\therefore$ Since the driving func., $\cos(3t)$ created $(s^2 + 9)$ in the denominator,

that generated $\frac{Ds + E}{s^2 + 9}$ term in Egn ①

$$\Rightarrow x_h(t) = \frac{9}{20} e^{-t} - \frac{7}{16} \cos(3t) + \frac{25}{16} \sin(3t) \quad \#$$

Week 8 : Laplace Transform II & Fourier Series I

9. Consider the function $f(x) = \begin{cases} k & \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} < |x| \leq \pi \end{cases};$



a) Find the Fourier series representation for f .
 ① $\frac{\pi}{2} \leq x \leq \pi$
 ② $-\pi \leq x \leq -\frac{\pi}{2}$

b) Using the result of problem 9, show that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

a) Rewriting $f(x)$

$$f(x) = \begin{cases} 0 & \text{for } -\pi \leq x < -\frac{\pi}{2} \\ k & \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

Since $f(x)$ is an even func., $b_n = 0$

a_0 = average value

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

Since $f(x)$ is symmetrical about y-axis,

$$a_0 = 2 \left[\frac{1}{2\pi} \int_0^{\pi} f(x) dx \right]$$

$$= \frac{1}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \left[\int_0^{\pi/2} k dx + \int_{\pi/2}^{\pi} 0 dx \right]$$

$$= \frac{k}{\pi} \int_0^{\pi/2} 1 dx$$

$$= \frac{k}{\pi} [x]_0^{\pi/2}$$

$$= \frac{k}{\pi} \left(\frac{\pi}{2} \right)$$

$$= \frac{k}{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$$

$$= \frac{2}{\pi} \left[\int_0^{\pi/2} k \cos(nx) dx + \int_{\pi/2}^{\pi} 0 dx \right]$$

$$= \frac{2k}{\pi} \int_0^{\pi/2} \cos(nx) dx$$

$$= \frac{2k}{\pi} \left[\frac{\sin(nx)}{n} \right]_0^{\pi/2}$$

$$= \frac{2k}{n\pi} \sin\left(\frac{n\pi}{2}\right) \text{ for } n = 1, 2, 3, \dots$$

Since $n = \text{even} \Rightarrow \sin\left(n \frac{\pi}{2}\right) = 0$

$$n = 1 \Rightarrow \sin\left(\frac{\pi}{2}\right) = 1$$

$$3 \Rightarrow \sin\left(\frac{3\pi}{2}\right) = -1$$

$$5 \Rightarrow \sin\left(\frac{5\pi}{2}\right) = 1$$

$$\left. \begin{array}{l} 1 \Rightarrow \sin\left(\frac{\pi}{2}\right) = 1 \\ 3 \Rightarrow \sin\left(\frac{3\pi}{2}\right) = -1 \\ 5 \Rightarrow \sin\left(\frac{5\pi}{2}\right) = 1 \end{array} \right\} (-1)^{m-1} \text{ for } m = 1, 2, 3, \dots$$

Odd numbers $\Rightarrow 2m - 1$ for $m = 1, 2, 3, \dots$

$$\therefore a_n = \frac{2k}{(2m-1)\pi} (-1)^{m-1} \cos[(2m-1)x]$$

where $n = 2m - 1$, for $m = 1, 2, 3, \dots$

$$\therefore f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx)$$

$$= \frac{k}{2} + \frac{2k}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m-1} \cos[(2m-1)x] \quad \#$$

b) Observing $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$,

Since RHS numerator always is 1, we make the cosine term 1 by setting $x=0$

Let $x=0$,

$$\begin{aligned} f(0) &= \frac{k}{2} + \frac{2k}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m-1} \quad (1) \\ &= \frac{k}{2} + \frac{2k}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right] \end{aligned}$$

Since $x=0$ falls in $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \Rightarrow f(0) = k$

$$\therefore k = \frac{k}{2} + \frac{2k}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right]$$

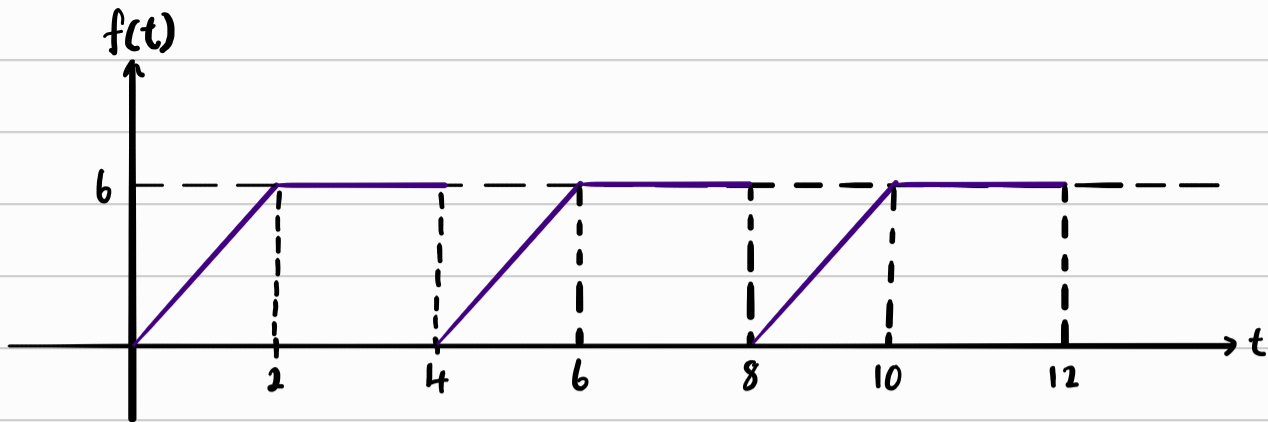
$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \quad [\text{Shown}] \quad \#$$

10. A periodic function $f(t)$, with period 4 units, is defined within the interval $0 \leq t < 4$ by

$$f(t) = \begin{cases} 3t & \text{if } 0 \leq t < 2 \\ 6 & \text{if } 2 \leq t < 4 \end{cases}$$

Sketch a graph of the function $f(t)$ for $0 \leq t < 12$ and obtain its Laplace transform.

Hint: See James, Theorem 5.5, pg. 409



By using standard formula for Laplace

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^{\infty} f(t) e^{-st} dt \\ &= \int_0^T f(t) e^{-st} dt + \int_T^{2T} f(t) e^{-st} dt + \int_{2T}^{3T} f(t) e^{-st} dt + \dots \end{aligned}$$

Notice each following pulse is just a time delay, T from the latest pulse.

Essentially,

$$f(t) = (\text{Ori pulse}) + (\text{Ori Pulse delayed by } T) + (\text{Ori Pulse delayed by } 2T) + \dots$$

In Laplace-land, a time-delay of T seconds is represented by multiplying e^{-sT}

$$\begin{aligned} \therefore \mathcal{L}\{f(t)\} &= \int_0^T f(t) e^{-st} dt + e^{-sT} \int_0^T f(t) e^{-st} dt + (e^{-sT})(e^{-sT}) \int_0^T f(t) e^{-st} dt + \dots \\ &= \int_0^T f(t) e^{-st} dt + e^{-sT} \int_0^T f(t) e^{-st} dt + e^{-2sT} \int_0^T f(t) e^{-st} dt + \dots \\ &= [1 + e^{-sT} + e^{-2sT} + e^{-3sT} + \dots] \cdot \int_0^T e^{-st} f(t) dt \end{aligned}$$

Using the infinite geometric series formula $[1 + r + r^2 + \dots = \frac{1}{1-r}]$

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt \\ &= \frac{1}{1 - e^{-4s}} \int_0^4 e^{-4s} f(t) dt, \text{ since } T = 4 \end{aligned}$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-4s}} \left[\int_0^2 (3t) e^{-st} dt + \int_2^4 (6) e^{-st} dt \right] \text{--- ①}$$

$$\begin{aligned} \int_0^2 (3t) e^{-st} dt &= \left[-\frac{3t}{s} e^{-st} - \frac{3}{s^2} e^{-st} \right]_0^2 \\ &= \left[\left(-\frac{6}{s} e^{-2s} - \frac{3}{s^2} e^{-2s} \right) - \left(0 - \frac{3}{s^2} \right) \right] \\ &= \frac{3}{s^2} - \frac{3}{s^2} e^{-2s} - \frac{6}{s} e^{-2s} \text{--- ②} \end{aligned}$$

D	I
+ 3t	e^{-st}
- 3	$-\frac{1}{s} e^{-st}$
+ 0	$\frac{1}{s^2} e^{-st}$

$$\begin{aligned} \int_2^4 6 e^{-st} dt &= 6 \left[-\frac{1}{s} e^{-st} \right]_2^4 \\ &= -\frac{6}{s} [e^{-4s} - e^{-2s}] \\ &= -\frac{6}{s} e^{-4s} + \frac{6}{s} e^{-2s} \text{--- ③} \end{aligned}$$

②, ③ → ①

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \frac{1}{1 - e^{-4s}} \left[\left(\frac{3}{s^2} - \frac{3}{s^2} e^{-2s} - \frac{6}{s} e^{-2s} \right) + \left(-\frac{6}{s} e^{-4s} + \frac{6}{s} e^{-2s} \right) \right] \\ F(s) &= \frac{1}{1 - e^{-4s}} \left[\frac{3}{s^2} - \frac{3}{s^2} e^{-2s} - \frac{6}{s} e^{-4s} \right] \\ &= \frac{1}{1 - e^{-4s}} \left[\frac{3}{s^2} (1 - e^{-2s}) - \frac{6}{s} e^{-4s} \right] \\ &= \frac{1}{s^2(1 - e^{-4s})} [3(1 - e^{-2s}) - 6s e^{-4s}] \text{--- ④} \end{aligned}$$

11. A periodic function $f(t)$ is an infinite train of unit impulses at $t = 0$ and repeated at intervals of $t = T$. Show that

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}}$$

The response of a harmonic oscillator to such a periodic stimulus is determined by the differential equation

$$\frac{d^2x}{dt^2} + \omega^2 x = f(t) \quad (t \geq 0)$$

Show that:

$$x(t) = \frac{1}{\omega} \sum_{n=0}^{\infty} u(t-nT) \sin \omega(t-nT), \quad (t \geq 0)$$

and sketch in MATLAB or otherwise, the responses from $t = 0$ to $t = 6\pi/\omega$ for the two cases (a) $T = \pi/\omega$ and (b) $T = 2\pi/\omega$.

Unit impulse can be represented with the Dirac Delta function, $\delta(t)$

$$\therefore f(t) = \delta(t) + \delta(t-T) + \delta(t-2T) + \delta(t-3T) + \dots$$

Taking Laplace Transform on both sides and using $\mathcal{L}\{\delta(t-a)\} = e^{-as}$

$$\Rightarrow \mathcal{L}\{f(t)\} = e^0 + e^{-sT} + e^{-2sT} + e^{-3sT} + \dots$$

$$F(s) = 1 + e^{-sT} + e^{-2sT} + e^{-3sT} + \dots$$

$$= \sum_{n=0}^{\infty} e^{-nsT}$$

Using infinite geometric series formula, $1 + r + r^2 + \dots = \frac{1}{1-r}$

$$\Rightarrow F(s) = \frac{1}{1-e^{-sT}} \quad \#$$

Given $\frac{d^2x}{dt^2} + \omega^2 x = f(t)$ [Assume $x(0) = 0$; $x'(0) = 0$]

$$[s^2 X(s) - s \cancel{x(0)} - \cancel{x'(0)}] + \omega^2 X(s) = F(s)$$

$$s^2 X(s) + \omega^2 X(s) = F(s)$$

$$X(s) = \frac{F(s)}{s^2 + \omega^2}$$

$$= \left[\sum_{n=0}^{\infty} e^{-nsT} \right] \frac{1}{s^2 + \omega^2}$$

$$X(s) = \sum_{n=0}^{\infty} (e^{-nsT} \cdot \frac{1}{s^2 + \omega^2})$$

Applying $\mathcal{L}^{-1}$ on both sides,

$$x(t) = \sum_{n=0}^{\infty} u(t-nT) \cdot \frac{1}{\omega} \sin(\omega(t-nT))$$

$$= \frac{1}{\omega} \sum_{n=0}^{\infty} u(t-nT) \sin(\omega(t-nT)) \quad \#$$

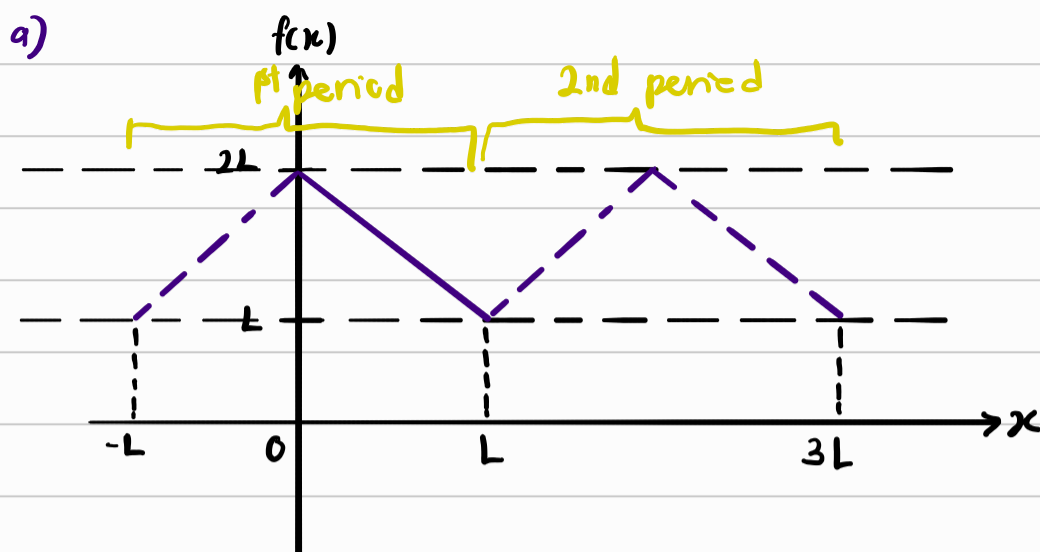
Multiplying $\frac{1}{s^2 + \omega^2}$ by e^{-nsT} in the s-domain delays a function by (nT) seconds. In the time domain, we represent this by multiplying the func. with unit step func., $u(t-nT)$ so that the function only "activates" after a certain delay

To make sure sine wave only activates when step function activates to avoid chopping into a wave midswing when the unit func. activates

Week 9 : Fourier Series II

5. Consider the function f defined on the interval $0 < x < L$: $f(x) = 2L - x$.

- Extend f as an even function in the interval $-L < x < L$ and then sketch the periodic extension of the resultant function over two periods.
- Find the constant term and a simplified form for the higher coefficients of the Fourier cosine series for $f(x)$. Write out explicitly the first four non-zero terms of the Fourier series.
- State the value of the sum of the series at $x = L$. Hence, or otherwise, find the sum of the series $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$



D	I
$2L - x$	$\cos\left(\frac{n\pi x}{L}\right)$
-1	$\frac{L}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$
0	$-\frac{L^2}{n^2\pi^2} \cos\left(\frac{n\pi x}{L}\right)$

b) Since we did an even half range expansion, $b_n = 0$

$$\begin{aligned}
 a_0 &= 2 \left[\frac{1}{2L} \int_0^L f(x) dx \right] \\
 &= \frac{1}{L} \int_0^L (2L - x) dx \\
 &= \frac{1}{L} \left[2Lx - \frac{x^2}{2} \right]_0^L \\
 &= \frac{1}{L} \left[(2L^2 - \frac{L^2}{2}) - 0 \right] \\
 &= \frac{3L}{2}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= 2 \left[\frac{1}{L} \int_0^L (2L - x) \cos\left(\frac{n\pi x}{L}\right) dx \right] \\
 &= \frac{2}{L} \int_0^L (2L - x) \cos\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{2}{L} \left[\frac{L(2L - x)}{n\pi} \sin\left(\frac{n\pi x}{L}\right) - \frac{L^2}{n^2\pi^2} \cos\left(\frac{n\pi x}{L}\right) \right]_0^L \\
 &= \frac{2}{L} \left[\left(0 - \frac{L^2}{n^2\pi^2} (-1)^n\right) - \left(0 - \frac{L^2}{n^2\pi^2} (1)\right) \right] \\
 &= \frac{2}{L} \left[\frac{L^2}{n^2\pi^2} (1 - (-1)^n) \right] \\
 &= \frac{2L}{n^2\pi^2} (1 - (-1)^n) \\
 &= \frac{4L}{n^2\pi^2}, \quad \begin{array}{l} \text{since } n = \text{even} \Rightarrow (1 - 1) = 0 \\ \quad \quad \quad n = \text{odd} \Rightarrow (1 + 1) = 2 \end{array}
 \end{aligned}$$

$$\begin{aligned}
 \therefore f(x) &= \frac{3L}{2} + \frac{4L}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \left(\frac{1}{n^2}\right) \cos\left(\frac{n\pi x}{L}\right) \\
 &= \frac{3L}{2} + \frac{4L}{\pi^2} \left[\cos\left(\frac{\pi x}{L}\right) + \frac{1}{3^2} \cos\left(\frac{3\pi x}{L}\right) + \frac{1}{5^2} \cos\left(\frac{5\pi x}{L}\right) + \dots \right] \neq
 \end{aligned}$$

c) When $x = L$,

$$f(L) = \frac{3L}{2} + \frac{4L}{\pi^2} \left[\cos(\pi) + \frac{1}{3^2} \cos(3\pi) + \frac{1}{5^2} \cos(5\pi) + \dots \right]$$

$$2L - L = \frac{3L}{2} + \frac{4L}{\pi^2} \left[-1 - \frac{1}{3^2} - \frac{1}{5^2} - \dots \right]$$

$$L = \frac{3L}{2} - \frac{4L}{\pi^2} \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

$$\left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] = \frac{\pi^2}{8} \quad \#$$

Week 10 : PDE I [Heat Egn]

9. Show there exists countably many values of ω for which the BVP

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0, \quad y(0) = 0, \quad \left. \frac{dy}{dx} \right|_{x=2} = -y(2)$$

has a non-trivial solution.

$$\text{Given } \frac{d^2 y}{dx^2} + \omega^2 y = 0 \quad \lambda^2 + \omega^2 = 0 \Rightarrow \lambda = \pm \sqrt{-\omega^2}$$

Case 1 : $\omega^2 > 0$

$$\text{Characteristic Egn : } \lambda^2 + \omega^2 = 0 \\ \lambda = \pm i\omega$$

$$\therefore \text{GS : } y(x) = A \cos(\omega x) + B \sin(\omega x)$$

Applying Boundary conditions,

$$y(0) = 0 \Rightarrow A = 0$$

$$\therefore y(x) = B \sin(\omega x)$$

$$y'(x) = B\omega \cos(\omega x)$$

$$y'(2) = -y(2)$$

$$B\omega \cos(2\omega) = -B \sin(2\omega)$$

$$\omega \cos(2\omega) = -\sin(2\omega), \text{ since } B \neq 0$$

$$\tan(2\omega) = -\omega$$

Case 2 : $\omega^2 = 0$

$$\frac{d^2 y}{dx^2} = 0$$

$$y(x) = Ax + B$$

Applying boundary conditions,

$$y(0) = 0 \Rightarrow B = 0$$

$$y'(2) = -y(2)$$

$$A = -[2A + 0]$$

$$3A = 0$$

$$A = 0$$

[Trivial]

$\therefore$ Since each branch of $\tan(2\omega)$ spans infinitely vertically and repeats horizontally periodically infinitely, a straight line $-\omega$ of infinite length guaranteed to intersect $\tan(2\omega)$ at infinite number of times.
 $\Rightarrow$ Having infinitely many non trivial solutions $\neq$

10. Show that the heat problem when modelling heat loss due to radioactive decay in a bar $\frac{\partial u}{\partial t} = \lambda \frac{\partial^2 u}{\partial x^2} + Ne^{-\alpha x}$ with $u(0, t) = u(L, t) = 0$ for all t , and $u(x, 0) = f(x)$, can be reduced to a problem for the homogeneous heat equation for $v(x, t)$ by setting $u(x, t) = v(x, t) + w(x)$, $v(0, t) = v(L, t) = 0$ for all t , and $v(x, 0) = f(x) - w(x)$. Find the $w(x)$ required to achieve this.

$$\text{Given } \begin{cases} \frac{\partial u}{\partial t} = \lambda \frac{\partial^2 u}{\partial x^2} + Ne^{-\alpha x} & \text{--- (1)} \\ u(x, t) = v(x, t) + w(x) & \text{--- (2)} \end{cases}$$

(2) $\rightarrow$ (1)

$$\begin{aligned} \left[\frac{\partial v}{\partial t} + 0 \right] &= \lambda \left[\frac{\partial^2 v}{\partial t^2} + w''(x) \right] + Ne^{-\alpha x} \\ \frac{\partial v}{\partial t} &= \lambda \frac{\partial^2 v}{\partial t^2} + \lambda w''(x) + Ne^{-\alpha x} \\ -\lambda \frac{\partial^2 v}{\partial t^2} + \frac{\partial v}{\partial t} &= \lambda w''(x) + Ne^{-\alpha x} \end{aligned}$$

For $v(x, t)$ to be a homogeneous equation, driving func. [RHS] = 0

$$\Rightarrow \lambda w''(x) + Ne^{-\alpha x} = 0$$

$$w''(x) = -\frac{N}{\lambda} e^{-\alpha x}$$

$$\begin{aligned} w'(x) &= -\frac{N}{\lambda} \int e^{-\alpha x} dx \\ &= \frac{N}{\alpha \lambda} e^{-\alpha x} + C_1 \end{aligned}$$

$$w(x) = -\frac{N}{\alpha^2 \lambda} e^{-\alpha x} + C_1 x + C_2 \quad \text{--- (3)}$$

Using boundary conditions,

$$\text{When } x = 0 \Rightarrow u(0, t) = 0 ; v(0, t) = 0$$

$$\therefore \text{Using Eqn (2)} \Rightarrow w(0) = 0$$

$$\text{When } x = L \Rightarrow u(L, t) = 0 ; v(L, t) = 0$$

$$\therefore \text{Using Eqn (2)} \Rightarrow w(L) = 0$$

Applying $w(0) = 0$ to Eqn ③

$$w(0) = -\frac{N}{\alpha^2 \lambda} e^0 + C_1(0) + C_2$$

$$0 = -\frac{N}{\alpha^2 \lambda} + C_2$$

$$C_2 = \frac{N}{\alpha^2 \lambda}$$

Applying $w(L) = 0$ to Eqn ③

$$w(L) = -\frac{N}{\alpha^2 \lambda} e^{-\alpha L} + C_1(L) + C_2$$

$$0 = -\frac{N}{\alpha^2 \lambda} e^{-\alpha L} + C_1 L + \frac{N}{\alpha^2 \lambda}$$

$$C_1 L = \frac{N}{\alpha^2 \lambda} [e^{-\alpha L} - 1]$$

$$C_1 = \frac{N}{L \alpha^2 \lambda} [e^{-\alpha L} - 1]$$

$$\begin{aligned} \therefore w(x) &= -\frac{N}{\alpha^2 \lambda} e^{-\alpha x} + \left[\frac{N}{L \alpha^2 \lambda} (e^{-\alpha L} - 1) \right] x + \frac{N}{\alpha^2 \lambda} \\ &= \frac{N}{\alpha^2 \lambda} \left[1 - e^{-\alpha x} + \frac{e^{-\alpha L} - 1}{L} x \right] \quad \# \end{aligned}$$

11. Assume that heat is transferred by conduction

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

in a thin rod of length 3 units. Assume that the ends of the rod are held at constant temperatures $u(0, t) = 2$ and $u(3, t) = 11$ for all $t > 0$, and initial temperature distribution is $u(x, 0) = 2 + x^2$. Assume the thermal diffusivity constant $\kappa = 13$.

The problem we have here is that the boundary conditions are non-homogeneous. In this question we will aim to solve this boundary value problem.

- (a) Let $u(x, t) = v(x, t) + \phi(x)$ where v is the **transient solution** and ϕ is the **steady state solution**. Substitute this form of u into the heat equation to find a **homogeneous** PDE in terms of v and an ODE in terms of ϕ .
- (b) If the purpose of writing $u(x, t) = v(x, t) + \phi(x)$ is to give a heat equation with homogeneous boundary conditions then write down the boundary conditions for the PDE of v and the boundary conditions for the ODE of ϕ .
- (c) Solve the boundary value problem for $\phi(x)$.
- (d) Then solve the boundary value problem for $v(x, t)$.
- (e) Hence, give the general solution of this heat conduction problem.

q) Given $\begin{cases} \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} & \text{--- (1)} \\ u(x, t) = v(x, t) + \phi(x) & \text{--- (2)} \end{cases}$

② $\rightarrow$ ①

$$\left[\frac{\partial v}{\partial t} + 0 \right] = k \left[\frac{\partial^2 v}{\partial x^2} + \phi''(x) \right]$$

$$\frac{\partial v}{\partial t} = k \frac{\partial^2 v}{\partial x^2} + k \phi''(x)$$

$$-k \frac{\partial^2 v}{\partial x^2} + \frac{\partial v}{\partial t} = k \phi''(x)$$

For $v(x, t)$ to be homogeneous, the driving func. [RHS] = 0

$$\Rightarrow k \phi''(x) = 0 \quad \text{[ODE]} \quad \text{--- (3)}$$

$$\therefore \frac{\partial v}{\partial t} = k \frac{\partial^2 v}{\partial x^2} \quad \text{[PDE]} \quad \text{--- (4)} \quad \#$$

b) Since we are going to solve the arbitrary constants of $v(x, t)$ in question d) with Fourier methods, we make sure the general solution is trigonometric.

$\Rightarrow$ Force $v(x, t)$ to have 2 roots

$$\Rightarrow v(0, t) = 0, \quad v(3, t) = 0 \quad \#$$

$$\begin{aligned} \text{When } x = 0 \\ \Rightarrow u(0, t) &= v(0, t) + \phi(0) \\ 2 &= 0 + \phi(0) \\ \phi(0) &= 2 \quad \# \end{aligned}$$

$$\begin{aligned} \text{When } x = 3 \\ u(3, t) &= v(3, t) + \phi(3) \\ 11 &= 0 + \phi(3) \\ \phi(3) &= 11 \quad \# \end{aligned}$$

$$\begin{aligned} \therefore v(0, t) &= 0, \quad \phi(0) = 2 \\ v(3, t) &= 0, \quad \phi(3) = 11 \quad \# \end{aligned}$$

c) Solving for Eqn (3)

$$\begin{aligned} \phi''(x) &= 0 \\ \phi(x) &= Ax + B \end{aligned}$$

Applying boundary conditions,

$$\phi(0) = 2 \Rightarrow B = 2$$

$$\phi(3) = 11 \Rightarrow 11 = 3A + 2$$

$$A = 3$$

$$\therefore \phi(x) = 3x + 2 \quad \#$$

d) Solving for (4)

$$\frac{\partial v}{\partial t} = 13 \frac{\partial^2 v}{\partial x^2} \quad - (5)$$

Assume $v(x, t) = X(x) T(t)$, - (6)

(6) $\rightarrow$ (5)

$$X(x) T'(t) = 13 X''(x) T(t)$$

$$\frac{T'(t)}{13 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

$$\therefore X'' + \lambda X = 0 \quad - (7)$$

$$T' + 13\lambda T = 0 \quad - (8)$$

Solving for (7)

Characteristic Eqn : $r^2 + \lambda = 0$
 $r = \pm i\sqrt{\lambda}$

Since there are 2 roots [$x=0, x=3$]

$$\text{Let } \lambda = \alpha^2 \Rightarrow r = \pm i\alpha$$

$$\therefore \text{GS : } X(x) = A \cos(\alpha x) + B \sin(\alpha x)$$

Applying Boundary Conditions

$$X(0) = 0 \Rightarrow A = 0$$

$$\therefore X(x) = B \sin(\alpha x)$$

$$\text{Since } X(3) = 0 \Rightarrow B \sin(3\alpha) = 0$$

$$B = 0 \text{ or } \sin(3\alpha) = 0$$

[Reject $\therefore$ Trivial]

$$3\alpha = n\pi$$

$$\alpha = \frac{n\pi}{3}, \text{ for } n=1, 2, 3, \dots$$

$$\therefore \lambda_n = \left(\frac{n\pi}{3}\right)^2$$

$$\therefore X_n(x) = \sin\left(\frac{n\pi x}{3}\right)$$

Boundaries

$$v(0, t) = 0 \Rightarrow X(0) = 0$$

$$v(3, t) = 0 \Rightarrow X(3) = 0$$

Solving for (8)

$$\frac{dT}{dt} + 13\left(\frac{n\pi}{3}\right)^2 T = 0$$

$$\frac{1}{T} \frac{dT}{dt} = -13\left(\frac{n\pi}{3}\right)^2$$

$$\int \frac{1}{T} dT = -13\left(\frac{n\pi}{3}\right)^2 \int 1 dt$$

$$\ln|T| = -13\left(\frac{n\pi}{3}\right)^2 t + C$$

$$T_n(t) = C_n e^{-13\left(\frac{n\pi}{3}\right)^2 t}$$

$$\therefore v_n(x, t) = X_n(x) \cdot T_n(t)$$

$$= B_n \sin\left(\frac{n\pi x}{3}\right) e^{-\frac{13n^2\pi^2 t}{9}}$$

$$\therefore v(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{3}\right) e^{-\frac{13n^2\pi^2 t}{9}}$$

When $t = 0$, using Eqn (2)

$$v(x, 0) = u(x, 0) = \phi(x)$$

$$= (2+x^2) - (3x+2)$$

$$= x^2 - 3x$$

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{3}\right) = x^2 - 3x$$

[Next Pg]

Using Fourier Formula

$$\begin{aligned}
 B_n &= \frac{2}{3} \int_0^3 (x^2 - 3x) \sin\left(\frac{n\pi x}{3}\right) dx \\
 &= \frac{2}{3} \left[-(x^2 - 3x) \frac{3}{n\pi} \cos\left(\frac{n\pi x}{3}\right) \right. \\
 &\quad \left. + (2x - 3) \frac{9}{n^2\pi^2} \sin\left(\frac{n\pi x}{3}\right) \right. \\
 &\quad \left. + \frac{54}{n^3\pi^3} \cos\left(\frac{n\pi x}{3}\right) \right]_0^3 \\
 &= \frac{2}{3} \left[\frac{54}{n^3\pi^3} (-1)^n - \frac{54}{n^3\pi^3} \right] \\
 &= \frac{36}{n^3\pi^3} ((-1)^n - 1) \\
 &= -\frac{72}{n^3\pi^3} \text{ for } n = 1, 3, 5, \dots
 \end{aligned}$$

D	I
+ $x^2 - 3x$	$\sin\left(\frac{n\pi x}{3}\right)$
- $2x - 3$	$-\frac{3}{n\pi} \cos\left(\frac{n\pi x}{3}\right)$
+ 2	$-\frac{9}{n^2\pi^2} \sin\left(\frac{n\pi x}{3}\right)$
- 0	$\frac{27}{n^3\pi^3} \cos\left(\frac{n\pi x}{3}\right)$

$$\therefore v(x, t) = \sum_{n=1,3,5,\dots}^{\infty} -\frac{72}{n^3\pi^3} \sin\left(\frac{n\pi x}{3}\right) e^{-\frac{13n^2\pi^2 t}{9}} \quad \#$$

$$\begin{aligned}
 \text{e) } u(x, t) &= \phi(x) + v(x, t) \\
 &= 3x + 2 + \sum_{n=1,3,5,\dots}^{\infty} -\frac{72}{n^3\pi^3} \sin\left(\frac{n\pi x}{3}\right) e^{-\frac{13n^2\pi^2 t}{9}} \quad \#
 \end{aligned}$$

Week 11 : PDE II [Laplace's Egn]

5. Poisson's equation occurs in many problems involving electric potential. Solve the Poisson equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -k, \quad (x, y) \in (0, \pi) \times (0, \infty)$$

for fixed constant $k > 0$, subject to the boundary conditions

$$\begin{aligned} &\text{homogeneous} \\ &\boxed{u(x, 0) = 0} \text{ for } 0 < x < \pi \\ &\boxed{u(0, y) = 0} \text{ for } y > 0 \\ &\boxed{u(\pi, y) = 1} \text{ for } y > 0 \\ &\text{non-homogeneous} \end{aligned}$$

Given $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -k$ — (1)

Assume $u(x, y) = X(x) Y(y)$ — (2)

(2) → (1)

$$\begin{aligned} X''(x) Y(y) + X(x) Y''(y) &= -k \\ \frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} &= -\frac{k}{X(x) Y(y)} \\ \frac{X''(x)}{X(x)} &= -\frac{Y''(y)}{Y(y)} - \frac{k}{X(x) Y(y)} \end{aligned}$$

Since it's impossible to separate the variables into LHS only depend on x , RHS only depend on y , assumption failed.

∴ We fall back to the basics where ^{Solve w/ separation of variables}

$u(x, y)$ = particular soln + homogeneous soln

$$u(x, y) = u_p(x) + u_h(x, y) \quad \text{— (3)}$$

(3) → (1)

$$\begin{aligned} \frac{\partial^2}{\partial x^2} [u_p + u_h] + \frac{\partial^2}{\partial y^2} [u_p + u_h] &= -k \\ \underbrace{\left[\frac{\partial^2 u_p}{\partial x^2} + \frac{\partial^2 u_p}{\partial y^2} \right]}_{\text{non-homo}} + \underbrace{\left[\frac{\partial^2 u_h}{\partial x^2} + \frac{\partial^2 u_h}{\partial y^2} \right]}_{\text{homo}} &= -k \\ \frac{\partial^2 u_p}{\partial x^2} + \left[\frac{\partial^2 u_h}{\partial x^2} + \frac{\partial^2 u_h}{\partial y^2} \right] &= -k \end{aligned}$$

We set the homogeneous part $= 0$,
 $\Rightarrow \frac{\partial^2 u_h}{\partial x^2} + \frac{\partial^2 u_h}{\partial y^2} = 0$ — (4)

$$\therefore \frac{\partial^2 u_p}{\partial x^2} + 0 = -k$$

$$\frac{\partial^2 u_p}{\partial x^2} = -k \text{ — (5)}$$

Boundary Interpretation

Since we are going to solve the arbitrary constants of $u_h(x, y)$ with Fourier methods, we make sure the general solution is trigonometric.

$\Rightarrow$ Force $u_h(x, y)$ to have 2 roots

$$\Rightarrow u_h(0, y) = 0, \quad u_h(\pi, y) = 0$$

Given $u(0, y) = 0$

$$u_p(0) + u_h(0, y) = 0$$

$$u_p(0) + 0 = 0$$

$$u_p(0) = 0$$

Given $u(\pi, y) = 1$

$$u_p(\pi) + u_h(\pi, y) = 1$$

$$u_p(\pi) + 0 = 1$$

$$u_p(\pi) = 1$$

Solving for (5)

$$u_p''(x) = -k$$

$$u_p'(x) = -kx + C_1$$

$$u_p(x) = -\frac{1}{2}kx^2 + C_1x + C_2$$

Applying boundary conditions

$$u_p(0) = 0 \Rightarrow C_2 = 0$$

$$\therefore u_p(x) = -\frac{1}{2}kx^2 + C_1x$$

Using $u_p(\pi) = 1$

$$-\frac{1}{2}k\pi^2 + \pi C_1 = 1$$

$$C_1 = \frac{1}{\pi} + \frac{k\pi}{2}$$

$$\therefore u_p(x) = -\frac{1}{2}kx^2 + \left[\frac{1}{\pi} + \frac{k\pi}{2}\right]x$$

$$= \frac{x}{\pi} + \frac{k}{2}[\pi x - x^2]$$

Solving for (4)

Assume $u_h(x, y) = X(x)Y(y)$ — (6)

(6) $\rightarrow$ (4)

$$X''Y + XY'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

$$\therefore X'' + \lambda X = 0 \text{ — (7)}$$

$$Y'' - \lambda Y = 0 \text{ — (8)}$$

Solving for ⑦

Characteristic Egn : $r^2 + \lambda = 0$
 $r = \pm \sqrt{-\lambda}$

Boundaries

$$u_n(0, y) = 0 \Rightarrow X(0) = 0$$

$$u_n(\pi, y) = 0 \Rightarrow X(\pi) = 0$$

Since there are 2 roots $[x=0, x=\pi]$, $\lambda > 0$

$$\text{Let } \lambda = \alpha^2 \Rightarrow r = \pm i\alpha$$

$$\therefore \text{GS : } X(x) = A \cos(\alpha x) + B \sin(\alpha x)$$

Applying Boundary Condition

$$X(0) = 0 \Rightarrow A = 0$$

$$\therefore X(x) = B \sin(\alpha x)$$

$$\text{Since } X(\pi) = 0$$

$$B \sin(\alpha \pi) = 0$$

$$B = 0 \text{ or } \sin(\alpha \pi) = 0$$

[Reject
 $\therefore$ trivial]

$$\alpha \pi = n\pi$$

$$\alpha = n \text{ for } n=1, 2, 3, \dots$$

$$\Rightarrow \lambda = n^2$$

$$\therefore X_n(x) = B \sin(nx)$$

Solving for ⑧

$$Y'' - n^2 Y = 0, \text{ since } \lambda = n^2$$

Characteristic Egn : $r^2 - n^2 = 0$

$$r = \pm n$$

$$\therefore \text{GS : } Y(y) = C e^{ny} + D e^{-ny}$$

$$\text{Since } y \rightarrow \infty, e^{ny} \rightarrow \infty$$

$$\Rightarrow \text{we set } C = 0 \text{ to avoid } Y(y) \rightarrow \infty$$

$$\therefore Y_n(y) = D e^{-ny}$$

$$\therefore u_{h_n}(x, y) = b_n \sin(nx) e^{-ny}$$

$$\therefore u_h(x, y) = \sum_{n=1}^{\infty} b_n \sin(nx) e^{-ny}$$

Since $u(x, 0) = 0$

$$u_p(x) + u_h(x, 0) = 0$$

$$u_h(x, 0) = -u_p(x)$$

$$\begin{aligned} \sum_{n=1}^{\infty} b_n \sin(nx) &= - \left[\frac{x}{\pi} + \frac{k}{2} [\pi x - x^2] \right] \\ &= \left[-\frac{x}{\pi} - \frac{k}{2} (\pi x - x^2) \right] \end{aligned}$$

D	I
$\frac{k}{2}x^2 - \left(\frac{1}{\pi} + \frac{k\pi}{2}\right)x$	$\sin(nx)$
$kx - \left(\frac{1}{\pi} + \frac{k\pi}{2}\right)$	$-\frac{1}{n} \cos(nx)$
k	$-\frac{1}{n^2} \sin(nx)$
0	$\frac{1}{n^3} \cos(nx)$

$$\begin{aligned} \therefore b_n &= \frac{2}{\pi} \int_0^{\pi} \left[-\frac{x}{\pi} - \frac{k}{2} (\pi x - x^2) \right] \sin(nx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} \left[\frac{k}{2} x^2 - \left(\frac{1}{\pi} + \frac{k\pi}{2}\right)x \right] \sin(nx) dx \\ &= \frac{2}{\pi} \left[\left[\frac{k}{2} x^2 - \left(\frac{1}{\pi} + \frac{k\pi}{2}\right)x \right] \left[-\frac{1}{n} \cos(nx) \right] \right. \\ &\quad \left. + \left[kx - \left(\frac{1}{\pi} + \frac{k\pi}{2}\right) \right] \left[-\frac{1}{n^2} \sin(nx) \right] \right. \\ &\quad \left. + \left[k \right] \left[\frac{1}{n^3} \cos(nx) \right] \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\left[(-1) \left(-\frac{1}{n} \cos(n\pi) \right) \right] + 0 + \left[\frac{k}{n^3} \cos(n\pi) - \frac{k}{n^3} \cos(0) \right] \right] \\ &= \frac{2}{\pi} \left[\frac{(-1)^n}{n} + \frac{k(-1)^n}{n^3} - \frac{k}{n^3} \right] \\ &= \frac{2}{\pi} \left[\frac{(-1)^n}{n} - \frac{k}{n^3} (1 - (-1)^n) \right] \\ &= \frac{2(-1)^n}{n\pi} - \frac{2k(1 - (-1)^n)}{\pi n^3} \end{aligned}$$

$$\therefore u(x, y) = \left[\frac{x}{\pi} + \frac{k}{2} (\pi x - x^2) \right] + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n\pi} - \frac{2k(1 - (-1)^n)}{\pi n^3} \right] \sin(nx) e^{-ny} \quad \#$$

6. Find the solution of Laplace's equation on an annulus region

$$\{(r, \theta) \in \mathbb{R}^2 : 0 \leq \theta \leq 2\pi, r_1 \leq r \leq r_2, r_1 > 0\}$$

with the boundary conditions

$$u(r_1, \theta) = u_0 \sin(\theta) \quad \text{and} \quad u(r_2, \theta) = 0 \quad \text{for } 0 \leq \theta \leq 2\pi$$

where $u_0 > 0$ is a constant.

Laplace Eqn in Polar : $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$ — ①

Assume $u(r, \theta) = R(r)T(\theta)$ — ②

② $\rightarrow$ ①

$$R''(r)T(\theta) + \frac{1}{r}R'(r)T(\theta) + \frac{1}{r^2}R(r)T''(\theta) = 0$$

Multiply $\frac{r^2}{R(r)T(\theta)}$ on both sides

$$\Rightarrow \frac{r^2 R'' + r R'}{R} + \frac{T''}{T} = 0$$

$$\frac{r^2 R'' + r R'}{R} = -\frac{T''}{T} = \lambda$$

$$\therefore r^2 R'' + r R' - \lambda R = 0 \text{ — ③}$$

$$T'' + \lambda T = 0 \text{ — ④}$$

Solving for ④

Characteristic Eqn : $r^2 + \lambda = 0$

$$r = \pm \sqrt{-\lambda}$$

Since its a annulus (ring) [Periodic]

$$\Rightarrow T(\theta) = T(\theta + 2\pi)$$

There are 2 valid cases

Case 1 : $\lambda > 0$

Case 2 : $\lambda = 0$

Case 1 : $\lambda > 0$

Solving for (4)

$$\text{Let } \lambda = \omega^2$$

$$\therefore \text{GS : } T(\theta) = A_n \cos(\omega\theta) + B_n \sin(\omega\theta)$$

$$\text{Since } T(\theta) = T(\theta + 2\pi)$$

$$\Rightarrow A \cos(\omega\theta) + B \sin(\omega\theta)$$

$$= A \cos(\omega\theta + \omega 2\pi) + B \sin(\omega\theta + \omega 2\pi)$$

For this to be true

$$\Rightarrow \omega = n, \text{ where } n = 1, 2, 3, \dots$$

$$\lambda = n^2$$

$\therefore$ General Solution for case 1

$$T_{n1}(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta)$$

Solving for (3)

$$r^2 R'' + r R' - n^2 R = 0$$

This is an Euler - Cauchy Egn,

$$R_n(r) = C_n r^n + D_n r^{-n}$$

$$\therefore u_{n1}(r, \theta) = [C_n r^n + D_n r^{-n}] [A_n \cos(n\theta) + B_n \sin(n\theta)]$$

$$= [(A_n r^n + B_n r^{-n}) \cos(n\theta) + (C_n r^n + D_n r^{-n}) \sin(n\theta)]$$

$$\therefore u_1(r, \theta) = \sum_{n=1}^{\infty} [(A_n r^n + B_n r^{-n}) \cos(n\theta) + (C_n r^n + D_n r^{-n}) \sin(n\theta)]$$

Case 2 : $\lambda = 0$

Solving for (4)

$$T'' = 0$$

$$T(\theta) = C_5 + C_6 \theta$$

$$\text{Since } T(\theta) = T(\theta + 2\pi)$$

$$\Rightarrow C_5 + C_6 \theta = C_5 + C_6 (\theta + 2\pi)$$

$$C_6 = 0$$

$$\therefore T(\theta) = C_5$$

Solving for (3)

$$r^2 R'' + r R' = 0$$

$$\frac{d}{dr} (r R') = 0$$

$$r R' = C_7$$

$$R' = C_7 / r$$

$$R(r) = C_7 \ln(r) + C_8$$

$$\therefore u_2(r, \theta) = [C_7 \ln(r) + C_8] [C_5]$$

$$= a_0 + b_0 \ln(r)$$

Total General solution, $u(r, \theta) = u_1(r, \theta) + u_2(r, \theta)$
 $\Rightarrow u(r, \theta) = [a_0 + b_0 \ln(r)] + \sum_{n=1}^{\infty} [(A_n r^n + B_n r^{-n}) \cos(n\theta) + (C_n r^n + D_n r^{-n}) \sin(n\theta)]$

Applying boundary condition

$$u(r_2, \theta) = 0$$

$$[a_0 + b_0 \ln(r_2)] + \sum_{n=1}^{\infty} [(A_n r_2^n + B_n r_2^{-n}) \cos(n\theta) + (C_n r_2^n + D_n r_2^{-n}) \sin(n\theta)] = 0$$

For infinite series $= 0$, all coefficients in the infinite series $= 0$

$$\left. \begin{aligned} \Rightarrow A_n r_2^n + B_n r_2^{-n} &= 0 \Rightarrow B_n = -A_n r_2^{2n} \\ C_n r_2^n + D_n r_2^{-n} &= 0 \Rightarrow D_n = -C_n r_2^{2n} \\ \text{and} \end{aligned} \right\} \textcircled{5}$$

$$a_0 + b_0 \ln(r_2) = 0 \Rightarrow a_0 = -b_0 \ln(r_2)$$

Sub ⑤ into $u(r, \theta)$

$$\begin{aligned} u(r, \theta) &= [-b_0 \ln(r_2) + b_0 \ln(r)] + \sum_{n=1}^{\infty} [(A_n r^n - A_n r_2^{2n} r^{-n}) \cos(n\theta) + (C_n r^n - C_n r_2^{2n} r^{-n}) \sin(n\theta)] \\ &= b_0 \ln\left(\frac{r}{r_2}\right) + \sum_{n=1}^{\infty} \left[A_n \left(r^n - \frac{r_2^{2n}}{r^n}\right) \cos(n\theta) + C_n \left(r^n - \frac{r_2^{2n}}{r^n}\right) \sin(n\theta) \right] \end{aligned}$$

Applying $u(r_1, \theta) = u_0 \sin(\theta)$

$$b_0 \ln\left(\frac{r_1}{r_2}\right) + \sum_{n=1}^{\infty} \left[A_n \left(r_1^n - \frac{r_2^{2n}}{r_1^n}\right) \cos(n\theta) + C_n \left(r_1^n - \frac{r_2^{2n}}{r_1^n}\right) \sin(n\theta) \right] = u_0 \sin(\theta)$$

By comparison, since RHS is $\sin(\theta)$, thus $n=1$

$$C_1 \left[r_1 - \frac{r_2^2}{r_1} \right] \sin \theta = u_0 \sin \theta$$

$$C_1 \left[\frac{r_1^2 - r_2^2}{r_1} \right] = u_0$$

$$C_1 = \frac{u_0 r_1}{r_1^2 - r_2^2}$$

$$\therefore u(r, \theta) = C_1 \left(r - \frac{r_2^2}{r} \right) \sin(n\theta)$$

$$= \left[\frac{u_0 r_1}{r_1^2 - r_2^2} \right] \left(r - \frac{r_2^2}{r} \right) \sin(\theta) \quad \#$$

Week 12 : PDE III [Advection / Wave Egn]

10. When a vibrating string is subject to an external vertical force that varies with the horizontal distance from the left end, the wave equation becomes

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + kx, \quad (x, t) \in (0, 1) \times (0, \infty)$$

for fixed constant $k > 0$. Solve this PDE subject to the conditions

$$u(0, t) = 0, \quad u(1, t) = 0 \text{ for } t > 0$$

$$u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \text{ for } 0 < x < 1$$

Rearranging $\Rightarrow \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = kx \quad \text{--- (1)}$

Since this is a non-homogeneous wave eqn, assuming $u(x, t) = X(x) \cdot T(t)$ won't work.

$\therefore$ We fall back to the basics where

$u(x, t) = \text{Particular solution} + \text{Homo. solution}$

$$u(x, t) = v(x) + w(x, t) \quad \text{--- (2)}$$

$\textcircled{2} \rightarrow \textcircled{1}$

$$\begin{aligned} \left[0 + \frac{d^2 w}{dt^2} \right] - c^2 \left[\frac{d^2 v}{dx^2} + \frac{d^2 w}{dx^2} \right] &= kx \\ \underbrace{\left[\frac{d^2 w}{dt^2} - c^2 \frac{d^2 w}{dx^2} \right]}_{\text{homo}} - \underbrace{c^2 \frac{d^2 v}{dx^2}}_{\text{non-homo}} &= kx \end{aligned}$$

We set homogeneous part = 0

$$\Rightarrow \frac{d^2 w}{dt^2} - c^2 \frac{d^2 w}{dx^2} = 0 \quad \text{--- (3)}$$

$$\therefore 0 - c^2 \frac{d^2 v}{dx^2} = kx$$

$$\frac{d^2 v}{dx^2} = -\frac{k}{c^2} x \quad \text{--- (4)}$$

Boundary Interpretation

Since we are going to solve the arbitrary constants of $w(x,y)$ with Fourier methods, we make sure the general solution is trigonometric.

$\Rightarrow$ Force $w(x,y)$ to have 2 roots

$$\Rightarrow w(0,t) = 0, \quad w(1,t) = 0$$

$$\text{Given } u(0,t) = 0$$

$$v(0) + w(0,t) = 0$$

$$v(0) + 0 = 0$$

$$v(0) = 0$$

$$\text{Given } u(1,t) = 0$$

$$v(1) + w(1,t) = 0$$

$$v(1) + 0 = 0$$

$$v(1) = 0$$

Solving for (4)

$$v''(x) = -\frac{k}{c^2}x$$

$$v'(x) = -\frac{k}{2c^2}x^2 + C_1$$

$$v(x) = -\frac{k}{6c^2}x^3 + C_1x + C_2$$

Applying Boundary conditions

$$v(0) = 0 \Rightarrow C_2 = 0$$

$$\therefore v(x) = -\frac{k}{6c^2}x^3 + C_1x$$

$$v(1) = 0 \Rightarrow C_1 = \frac{k}{6c^2}$$

$$\therefore v(x) = -\frac{k}{6c^2}x^3 + \frac{k}{6c^2}x$$

$$v(x) = \frac{k}{6c^2}[x - x^3]$$

Solving for (3)

$$\frac{d^2w}{dt^2} - c^2 \frac{d^2w}{dx^2} = 0$$

$$\text{Assume } w(x,t) = X(x)T(t) \text{ — (5)}$$

Sub (5) $\rightarrow$ (3)

$$X T'' - c^2 [X''(x) T(t)] = 0$$

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = -\lambda$$

$$\therefore X'' + \lambda X = 0 \text{ — (6)}$$

$$T'' + \lambda c^2 T = 0 \text{ — (7)}$$

Solving for ⑥

$$X'' + \lambda X = 0$$

Characteristic Eqn : $r^2 + \lambda = 0$

$$r = \pm \sqrt{-\lambda}$$

Boundaries

$$w(0, t) = 0 \Rightarrow X(0) = 0$$

$$w(1, t) = 0 \Rightarrow X(1) = 0$$

Since there are 2 roots $[x=0, x=1] \Rightarrow \lambda > 0$

$$\text{Let } \lambda = \mu^2 \Rightarrow r = \pm i\mu$$

$$\therefore \text{GS : } X(x) = C_1 \cos(\mu x) + C_2 \sin(\mu x)$$

Applying boundary conditions

$$X(0) = 0 \Rightarrow C_1 = 0$$

$$\therefore X(x) = C_2 \sin(\mu x)$$

$$X(1) = 0 \Rightarrow C_2 \sin(\mu) = 0$$

$$C_2 = 0 \quad \text{or} \quad \sin(\mu) = 0$$

[$\therefore$ Trivial]
Reject

$$\mu = n\pi, \text{ for } n=1, 2, 3, \dots$$

$$\lambda = (n\pi)^2$$

$$\therefore X_n(x) = C_2 \sin(n\pi x)$$

Solving for ⑦

$$T'' + (n\pi)^2 c^2 T = 0, \text{ using } \lambda = (n\pi)^2$$

Characteristic eqn : $r^2 + (n\pi)^2 c^2 = 0$

$$r = \pm i(n\pi c)$$

$$\therefore \text{GS : } T_n(t) = A_n \cos(n\pi c t) + B_n \sin(n\pi c t) \text{ --- } \textcircled{8}$$

Boundaries

$$\text{Given } \frac{dy}{dt} \Big|_{t=0} = 0$$

$$X(x) \cdot T'(0) = 0$$

$$T'(0) = 0$$

Differentiating Egn (8)

$$T'_n(t) = -A_n(n\pi c) \sin(n\pi ct) + B_n(n\pi c) \cos(n\pi ct)$$

$$\text{Using } T'(0) = 0,$$

$$0 + B_n(n\pi c)(1) = 0$$

$$B_n = 0$$

$$\therefore T_n(t) = A_n \cos(n\pi ct)$$

$$\therefore w_n(x, t) = X_n(x) \cdot T_n(t)$$

$$= A_n \sin(n\pi x) \cos(n\pi ct)$$

$$\therefore w(x, t) = \sum_{n=1}^{\infty} A_n \sin(n\pi x) \cos(n\pi ct)$$

$$\text{Given } u(x, 0) = 0$$

$$v(x) + w(x, 0) = 0$$

$$w(x, 0) = -v(x)$$

$$w(x, 0) = \frac{k}{6c^2} (x^3 - x)$$

$$\sum_{n=1}^{\infty} A_n \sin(n\pi x) = \frac{k}{6c^2} (x^3 - x)$$

Using Fourier formula,

$$A_n = \frac{2}{1} \int_0^1 \frac{k}{6c^2} (x^3 - x) \sin(n\pi x) dx$$

$$= \frac{k}{3c^2} \int_0^1 (x^3 - x) \sin(n\pi x) dx$$

$$\begin{aligned}
 A_n &= \frac{k}{3c^2} \left[(x^3 - x) \left(-\frac{1}{n\pi} \cos(n\pi x) \right) \right. \\
 &\quad - (3x^2 - 1) \left(-\frac{1}{(n\pi)^2} \sin(n\pi x) \right) \\
 &\quad + (6x) \left(\frac{1}{(n\pi)^3} \cos(n\pi x) \right) \\
 &\quad \left. - (6) \left(\frac{1}{(n\pi)^4} \sin(n\pi x) \right) \right]_0^1 \\
 &= \frac{k}{3c^2} \left[(0 - 0) \right. \\
 &\quad - (0) \\
 &\quad + \left(\frac{6}{n^3 \pi^3} \cos(n\pi) \right) \\
 &\quad \left. - (0) \right] \\
 &= \frac{k}{3c^2} \left[\frac{6}{n^3 \pi^3} (-1)^n \right] \\
 &= \frac{2k(-1)^n}{c^2 n^3 \pi^3}
 \end{aligned}$$

	D	I
+	$x^3 - x$	$\sin(n\pi x)$
-	$3x^2 - 1$	$-\frac{1}{n\pi} \cos(n\pi x)$
+	$6x$	$-\frac{1}{(n\pi)^2} \sin(n\pi x)$
-	6	$\frac{1}{(n\pi)^3} \cos(n\pi x)$
+	0	$\frac{1}{(n\pi)^4} \sin(n\pi x)$

$$\therefore w(x, t) = \sum_{n=1}^{\infty} \left[\frac{2k(-1)^n}{c^2 n^3 \pi^3} \right] \sin(n\pi x) \cos(n\pi ct)$$

$$\begin{aligned}
 \therefore u(x, t) &= v(x) + w(x, t) \\
 &= \frac{k}{6c^2} (x - x^3) + \sum_{n=1}^{\infty} \left[\frac{2k(-1)^n}{c^2 n^3 \pi^3} \right] \sin(n\pi x) \cos(n\pi ct) \quad \#
 \end{aligned}$$

11. Consider a thin beam of length 1 unit, with no external body forces applied ($f(x, t) = 0$) and define $\alpha^2 = \frac{\rho}{EI}$ then the beam equation becomes

$$\frac{\partial^4 u}{\partial x^4} + \alpha^2 \frac{\partial^2 u}{\partial t^2} = 0, \quad (x, t) \in (0, 1) \times (0, \infty)$$

Solve the beam equation given that the ends of the beam are simply supported

$$u(0, t) = 0, \quad \left. \frac{\partial^2 u}{\partial x^2} \right|_{x=0} = 0 \text{ for } t > 0$$

$$u(1, t) = 0, \quad \left. \frac{\partial^2 u}{\partial x^2} \right|_{x=1} = 0 \text{ for } t > 0$$

and has initial conditions

$$u(x, 0) = f(x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x) \text{ for } 0 < x < 1$$

where

(a) $f(x) = \sin(\pi x)$ and $g(x) = \sin(\pi x)$

(b) $f(x) = x(1 - x)$ and $g(x) = 0$

Given $\frac{\partial^4 u}{\partial x^4} + \alpha^2 \frac{\partial^2 u}{\partial t^2} = 0$ — ①

Assume $u(x, t) = X(x) T(t)$ — ②

② → ①

$$X''''(x) T(t) + \alpha^2 X(x) T''(t) = 0$$

$$\frac{X''''}{X} = -\alpha^2 \frac{T''}{T} = \lambda^4$$

$$\therefore X'''' - \lambda^4 X = 0 \text{ — ③}$$

$$T'' + \frac{1}{\alpha^2} \lambda^4 T = 0 \text{ — ④}$$

Solving for ③

Characteristic Eqn: $r^4 - \lambda^4 = 0$

$$(r^2 + \lambda^2)(r^2 - \lambda^2) = 0$$

$$r = \pm \lambda, \quad r = \pm i\lambda$$

Boundaries

$$u(0, t) = 0 \Rightarrow X(0) = 0$$

$$u(1, t) = 0 \Rightarrow X(1) = 0$$

$$u_{xx}(0, t) = 0 \Rightarrow X''(0) = 0$$

$$u_{xx}(1, t) = 0 \Rightarrow X''(1) = 0$$

Due to superposition, the general soln will just be the sum of the general soln of each corresponding pair.

$$\therefore X(x) = [A \cos(\lambda x) + B \sin(\lambda x)] + [C \cosh(\lambda x) + D \sinh(\lambda x)]$$

$$X''(x) = \lambda^2 [-A \cos(\lambda x) - B \sin(\lambda x) + C \cosh(\lambda x) + D \sinh(\lambda x)]$$

Applying boundary conditions

Left Boundaries

$$\left. \begin{array}{l} X(0) = 0 \Rightarrow A + C = 0 \\ X''(0) = 0 \Rightarrow -A + C = 0 \end{array} \right\} \begin{array}{l} A = 0 \\ C = 0 \end{array}$$

$$\therefore X(x) = B \sin(\lambda x) + D \sinh(\lambda x)$$

Right Boundaries

$$X(1) = 0 \Rightarrow B \sin(\lambda) + D \sinh(\lambda) = 0 \quad \text{--- (5)}$$

$$X''(1) = 0 \Rightarrow -B \sin(\lambda) + D \sinh(\lambda) = 0 \quad \text{--- (6)}$$

$$\textcircled{5} + \textcircled{6} \Rightarrow 2D \sinh(\lambda) = 0$$

$$D = 0 \text{ or } \sinh(\lambda) = 0$$

[Invalid]

$$\therefore B \sin(\lambda) = 0$$

$$B = 0 \text{ or } \sin(\lambda) = 0$$

$$\left[\begin{array}{l} \text{Reject} \\ \therefore \text{Trivial} \end{array} \right] \quad \lambda = n\pi \text{ for } n=1, 2, 3, \dots$$

$$\therefore X_n(x) = \sin(n\pi x)$$

Solving for (4)

$$T'' + \frac{(n\pi)^4}{\alpha^2} T = 0$$

$$\text{Let } \omega_n = \frac{(n\pi)^2}{\alpha},$$

$$\Rightarrow T'' + \omega_n^2 T_n = 0$$

Characteristic Eqn : $r^2 + \omega_n^2 = 0$

$$r = \pm i\omega_n$$

$$\therefore GS : T_n(t) = a_n \cos(\omega_n t) + b_n \sin(\omega_n t)$$

$$\begin{aligned}\therefore u_n(x, t) &= X_n(x) \cdot T_n(x) \\ &= \sin(n\pi x) [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)]\end{aligned}$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \sin(n\pi x) [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)]$$

a) $f(x) = \sin(\pi x)$ and $g(x) = \sin(\pi x)$

Boundary

$$u(x, 0) = f(x)$$

$$u_t(0, t) = g(x)$$

Given $u(x, 0) = f(x)$

$$\sum_{n=1}^{\infty} a_n \sin(n\pi x) = \sin(\pi x)$$

By comparison, $n = 1$

$$\Rightarrow a_1 \sin(\pi x) = \sin(\pi x)$$

$$a_1 = 1, a_{n \neq 1} = 0$$

Given $u_t(0, t) = g(x)$

$$\sum_{n=1}^{\infty} b_n \omega_n \sin(n\pi x) = \sin(\pi x)$$

By comparison, $n = 1$

$$\Rightarrow b_1 \omega_1 \sin(\pi x) = \sin(\pi x)$$

$$b_1 = \frac{1}{\omega_1}$$

$$= \frac{1}{(\frac{\pi^2}{\alpha})}, \text{ using } \omega_n = \frac{(n\pi)^2}{\alpha}$$

$$b_1 = \frac{\alpha}{\pi^2}, b_{n \neq 1} = 0$$

$$\therefore u(x, t) = \sin(\pi x) [a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t)]$$

$$= \sin(\pi x) \left[\cos\left(\frac{\pi^2}{\alpha} t\right) + \frac{\alpha}{\pi^2} \sin\left(\frac{\pi^2}{\alpha} t\right) \right] \quad \#$$

b) $f(x) = x(1-x)$ and $g(x) = 0$

Given $u_t(0, t) = g(x)$

$$\Rightarrow \sum_{n=1}^{\infty} b_n \omega_n \sin(n\pi x) = 0$$

$$\Rightarrow b_n = 0$$

Given $u(x, 0) = f(x)$

$$\sum_{n=1}^{\infty} a_n \sin(n\pi x) = x - x^2$$

Using Fourier

$$\begin{aligned} a_n &= \frac{2}{1} \int_0^1 (x - x^2) \sin(n\pi x) dx \\ &= 2 \left[(x - x^2) \left(-\frac{1}{n\pi} \cos(n\pi x) \right) \right. \\ &\quad \left. - (1-2x) \left(-\frac{1}{(n\pi)^2} \sin(n\pi x) \right) \right. \\ &\quad \left. + (-2) \left(\frac{1}{(n\pi)^3} \cos(n\pi x) \right) \right]_0^1 \end{aligned}$$

$$= 2 \int_0^1 0$$

$$= 0$$

$$+ \left(-\frac{2}{(n\pi)^3} \cos(n\pi) + \frac{2}{(n\pi)^3} \cos(0) \right)$$

$$= 2 \left[-\frac{2(-1)^n}{(n\pi)^3} + \frac{2}{(n\pi)^3} \right]$$

$$= 2 \left[\frac{2}{(n\pi)^3} (1 - (-1)^n) \right]$$

$$a_n = \frac{4(1 - (-1)^n)}{n^3 \pi^3}$$

	D	I
+	$x - x^2$	$\sin(n\pi x)$
-	$1 - 2x$	$-\frac{1}{n\pi} \cos(n\pi x)$
+	-2	$-\frac{1}{(n\pi)^2} \sin(n\pi x)$
-	0	$\frac{1}{(n\pi)^3} \cos(n\pi x)$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \sin(n\pi x) \left[a_n \cos(\omega_n t) + \overset{0}{b_n} \sin(\omega_n t) \right]$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin(n\pi x) \left[\frac{4(1 - (-1)^n)}{n^3 \pi^3} \right] \cos\left(\frac{n^2 \pi^2}{\alpha} t\right) \quad \neq$$